

Submission of project

Title: Secure File Storage and Access Management for Project Teams

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Date: 16/02/2026

Objective

The objective of this project is to design and implement a **secure file storage and access management system** for finance teams at *Globex Financial*. The system focuses on enforcing **user access controls, role-based permissions, Access Control Lists (ACLs), command history management, and security violation logging**.

The project aims to prevent unauthorized file access and modification while ensuring accountability, traceability, and compliance with audit requirements.

Implementation

To achieve the project objectives, a **Linux-based security solution** was implemented using built-in system administration and security tools. The solution was divided into multiple structured phases to ensure clarity, security, and audit readiness.

1. User and Group Management

Separate **project-based groups** were created to represent different teams (Project A and Project B). Each user was assigned to their respective group, ensuring **role-based access control (RBAC)**.

This segregation ensured that users only had access to resources relevant to their assigned project.

2. Secure Project Directory Configuration

A dedicated project directory was created to store sensitive files.

- Group ownership was assigned carefully.
- Directory permissions were configured to restrict access to authorized users only.
- The **sticky bit** was enabled to prevent users from deleting files created by others.

This setup ensured **data integrity** and **controlled collaboration**.

3. Access Control Lists (ACLs) Implementation

ACLs were configured to provide **fine-grained permission control** beyond standard Linux permissions.

- Specific users were granted read, write, and execute permissions.
- Unauthorized users and “others” were restricted from modifying or deleting files.
- Default ACLs ensured that newly created files inherited secure permissions automatically.

This prevented accidental or malicious changes to sensitive project data.

4. Command History Management

Command history limits were configured for selected users by modifying shell configuration files.

- The number of stored commands was restricted.
- This reduced exposure of sensitive command usage while still maintaining accountability.

This step enhanced **operational security** and reduced the risk of misuse.

5. Security Violation Logging Using Auditd

The Linux **auditd** service was installed and configured to monitor all read, write, execute, and attribute changes in the project directory.

- Custom audit rules were created to track access attempts.
- Logs captured both authorized and unauthorized activities.
- Audit logs were searchable using audit keys.

This provided **real-time monitoring** and **forensic evidence** for security audits.

6. Web-Based Audit Report System

To improve visibility and usability, audit logs were exported to a **web-accessible report system** using Apache.

- Logs were periodically updated using cron jobs.
- Access to audit reports was protected using HTTP authentication.

This allowed administrators to review security events easily while maintaining restricted access.

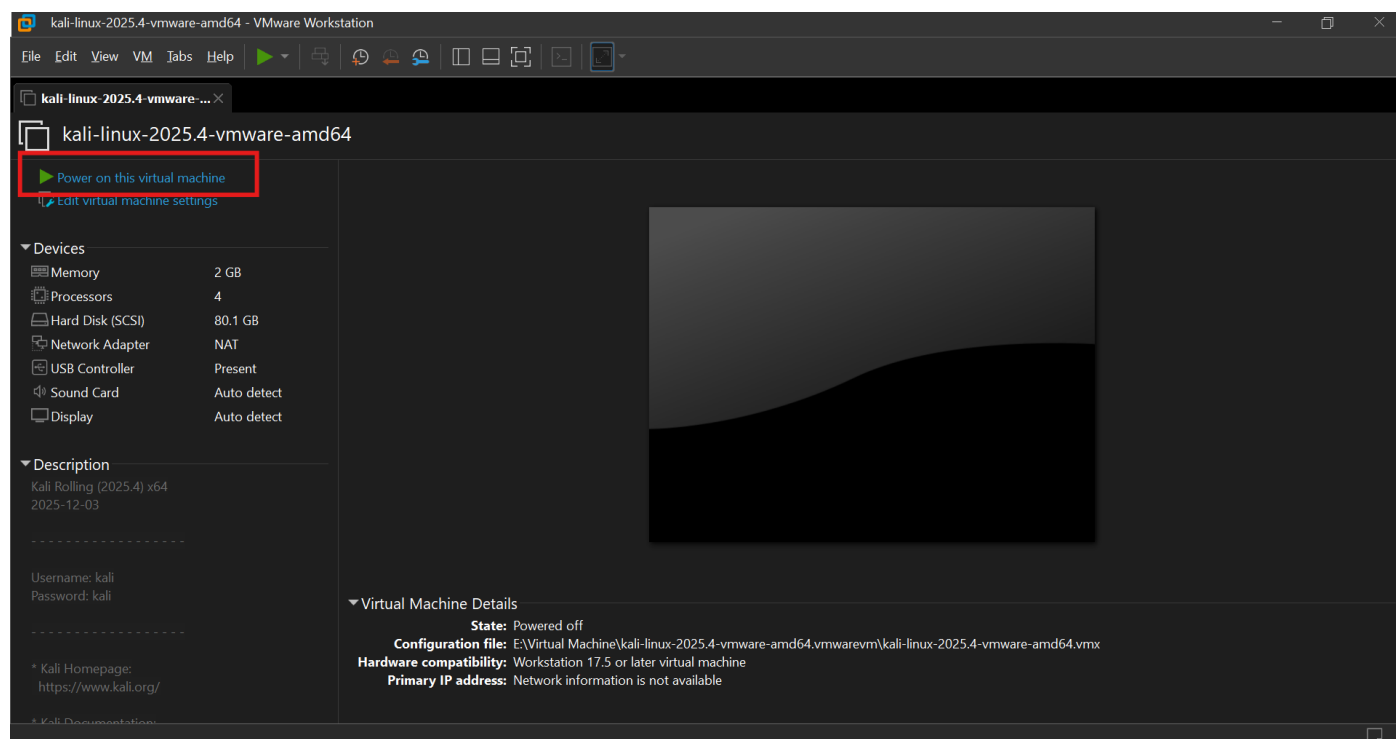
7. Testing and Validation

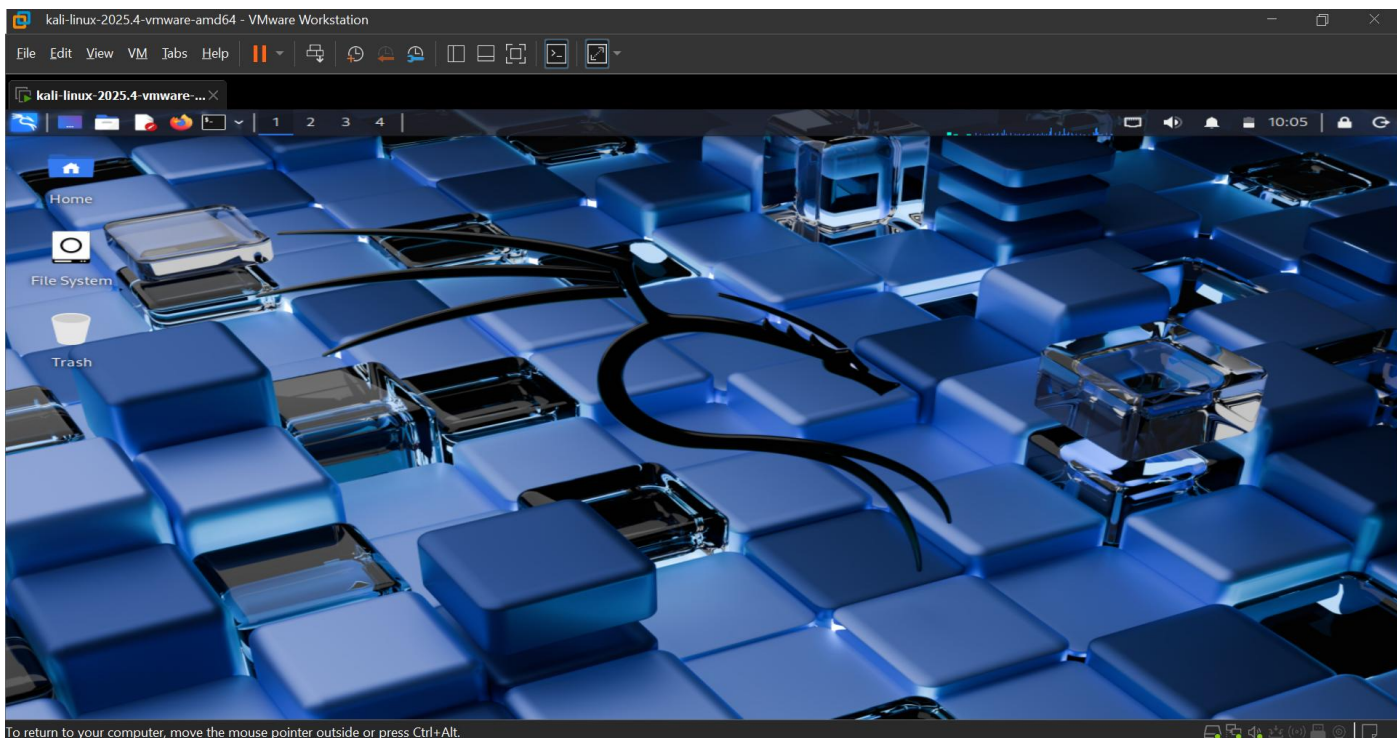
Multiple tests were conducted to validate the system:

- Authorized users successfully created and accessed files.
- Unauthorized users were denied access.
- File deletion restrictions worked as expected.
- Command history limits were enforced correctly.
- Security events were logged and visible in audit reports.

All tests confirmed that the system worked as intended.

Step 1: Powering the machine and login using credentials

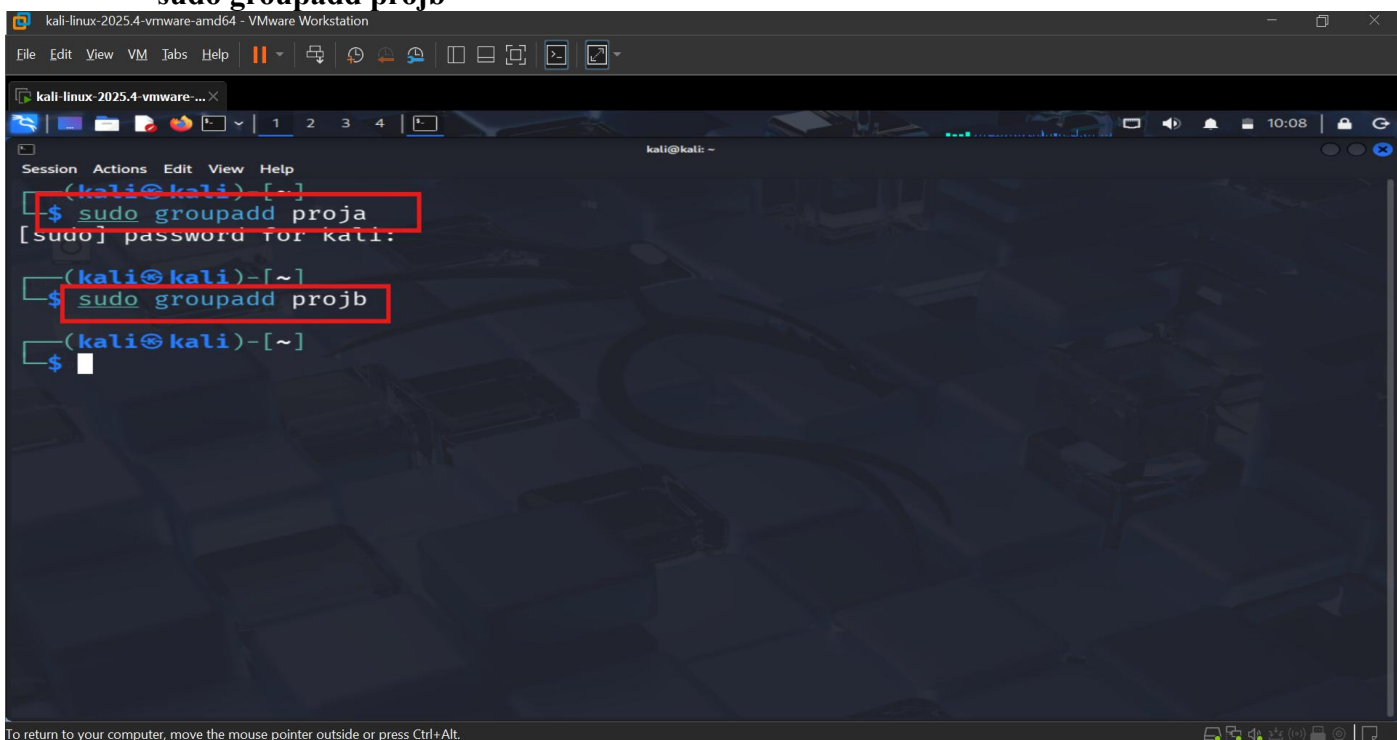




Step 2: Configuring “user accounts” and give permissions

2.1 Create Groups for Project Teams

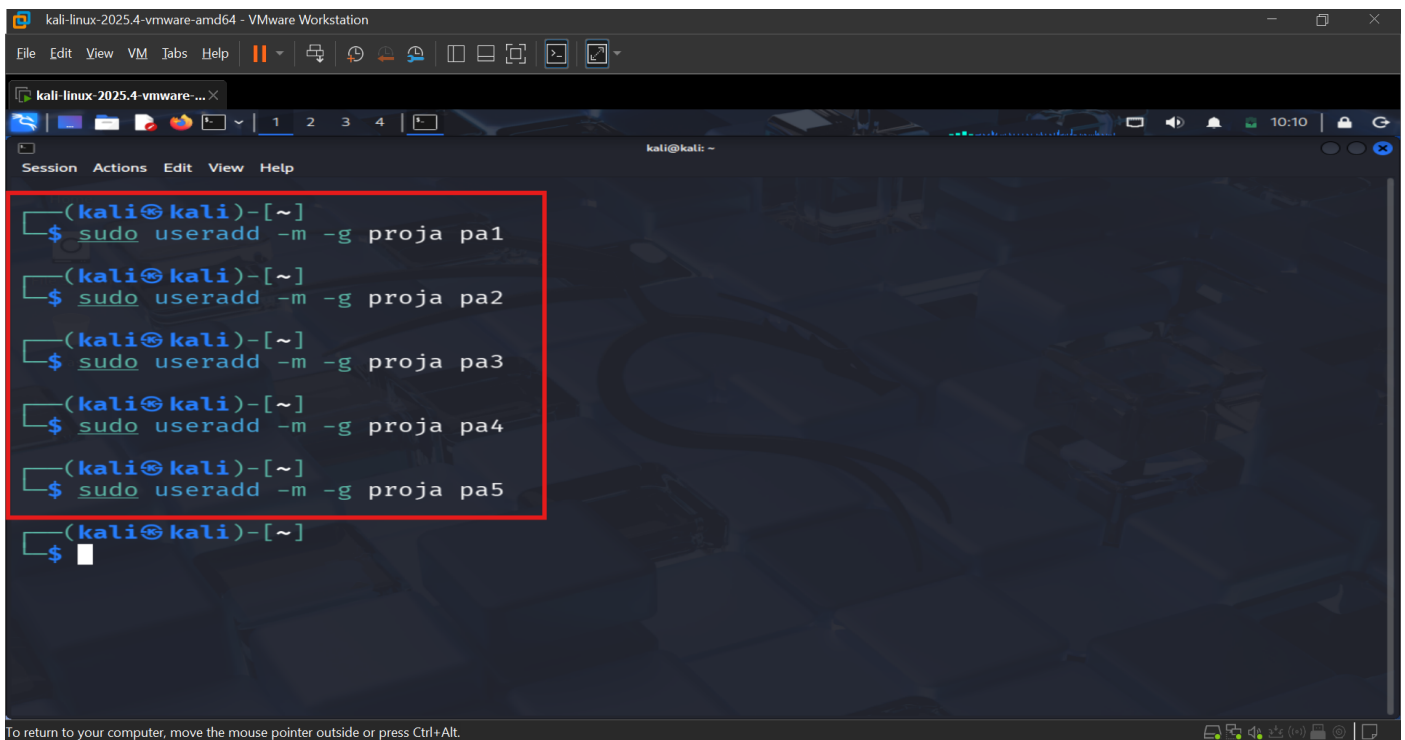
Command: **sudo groupadd proja**
sudo groupadd projb



2.2 Create User Accounts and Assign to Groups

For Project-A Team (proja users):

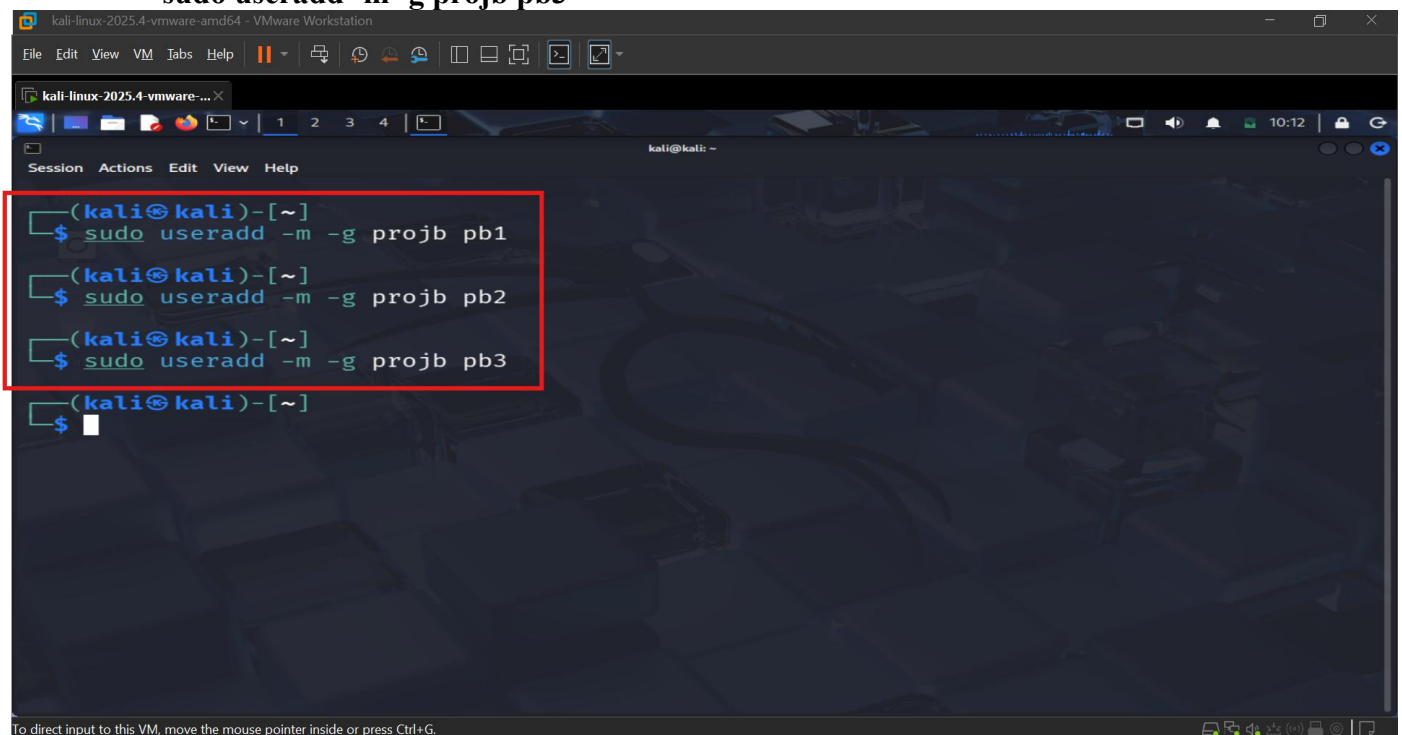
Command: **sudo useradd -m -g proja pa1**
sudo useradd -m -g proja pa2
sudo useradd -m -g proja pa3
sudo useradd -m -g proja pa4
sudo useradd -m -g proja pa5



```
kali@kali: ~  
$ sudo useradd -m -g proj a pa1  
$ sudo useradd -m -g proj a pa2  
$ sudo useradd -m -g proj a pa3  
$ sudo useradd -m -g proj a pa4  
$ sudo useradd -m -g proj a pa5  
$
```

For Project-B Team (projb users):

Command: `sudo useradd -m -g projb pb1`
`sudo useradd -m -g projb pb2`
`sudo useradd -m -g projb pb3`



```
kali@kali: ~  
$ sudo useradd -m -g projb pb1  
$ sudo useradd -m -g projb pb2  
$ sudo useradd -m -g projb pb3  
$
```

2.3 Set Passwords for Users (pa1,pa2,pa3,pa4,pa5)

Command: `sudo passwd pa1`
`sudo passwd pa2`
`sudo passwd pa3`
`sudo passwd pa4`
`sudo passwd pa5`


```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-... X
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo passwd pa1
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$ sudo passwd pa2
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$ sudo passwd pa3
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$ sudo passwd pa4
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$ sudo passwd pa5
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$
```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

Set Passwords for Users(pb1, pb2, pb3)

Command: **sudo passwd pb1**
sudo passwd pb2
sudo passwd pb3

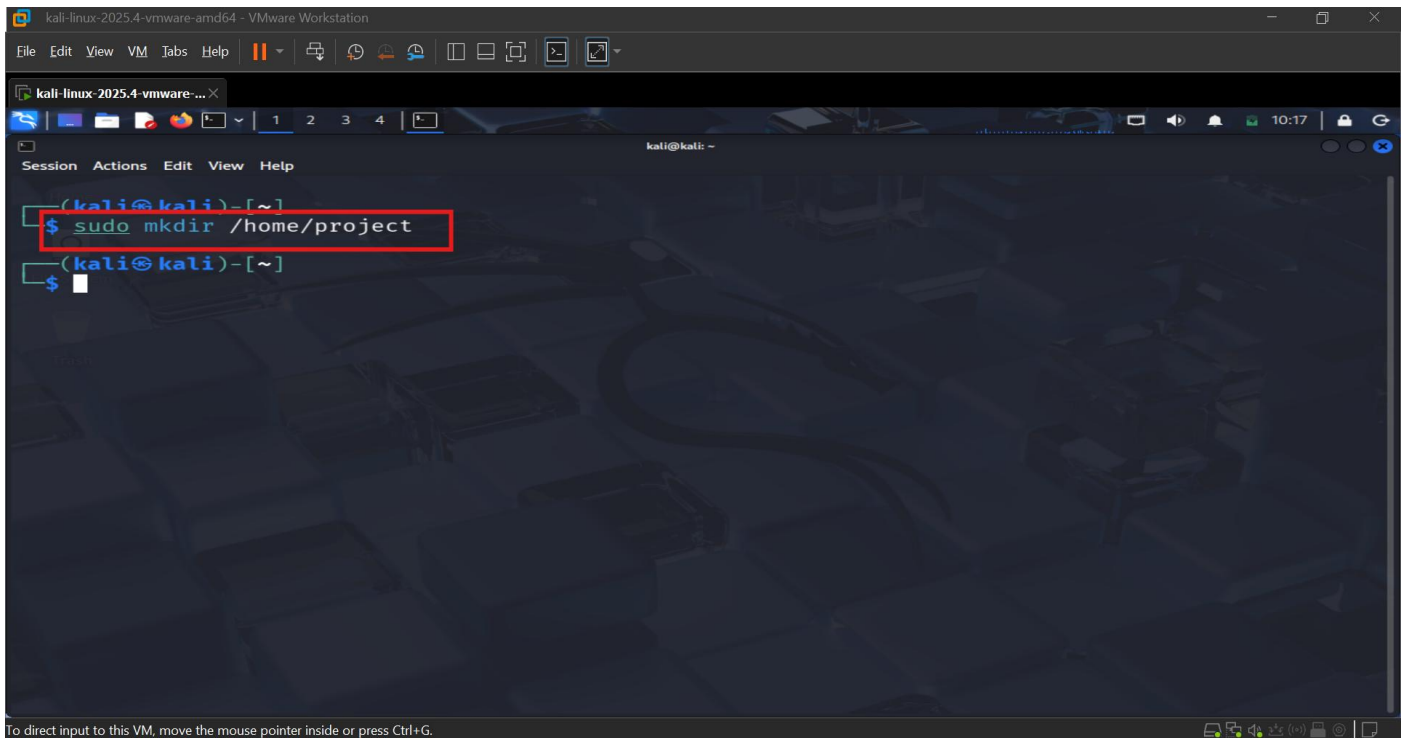
```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-... X
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo passwd pb1
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$ sudo passwd pb2
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$ sudo passwd pb3
New password:
Retype new password:
passwd: password updated successfully
(kali@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2.4 Create and Secure the Project Directory

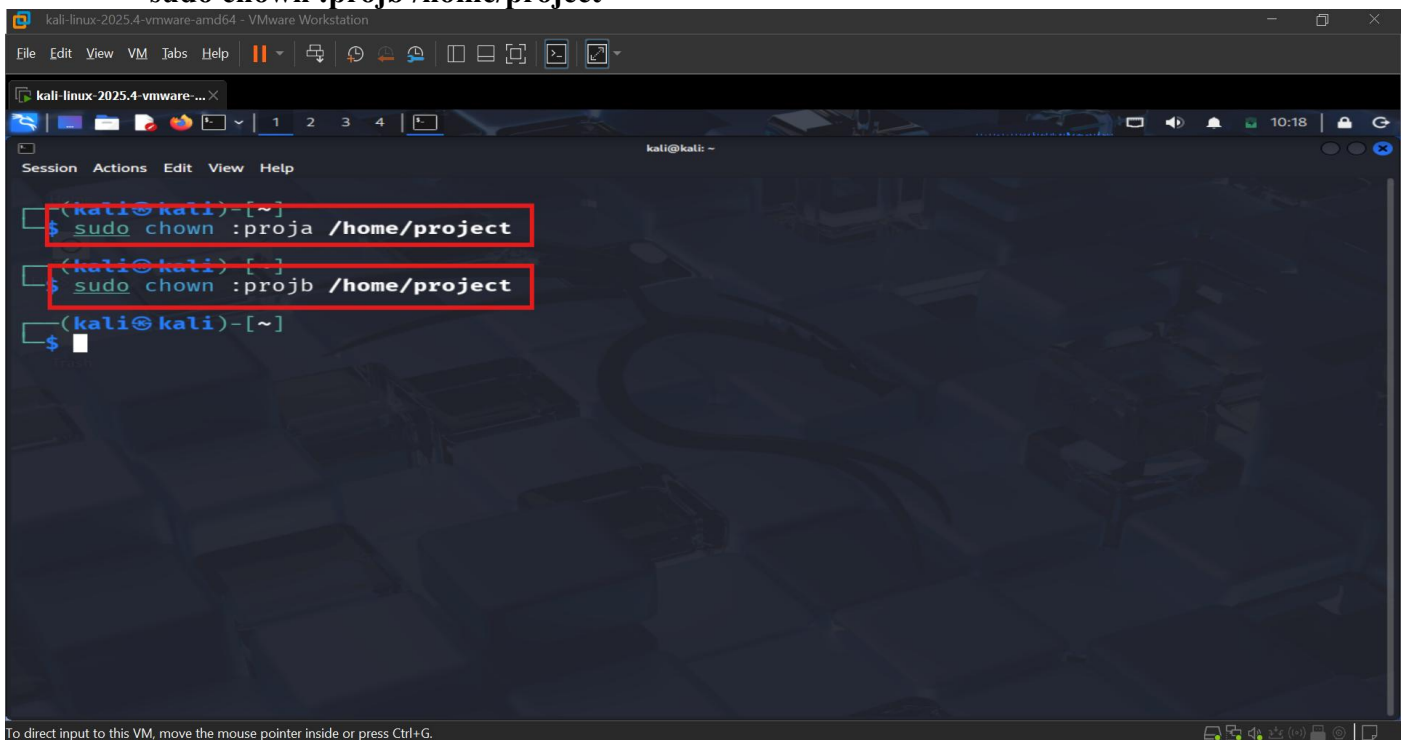
Create the Project Directory

Command: **sudo mkdir /home/project**



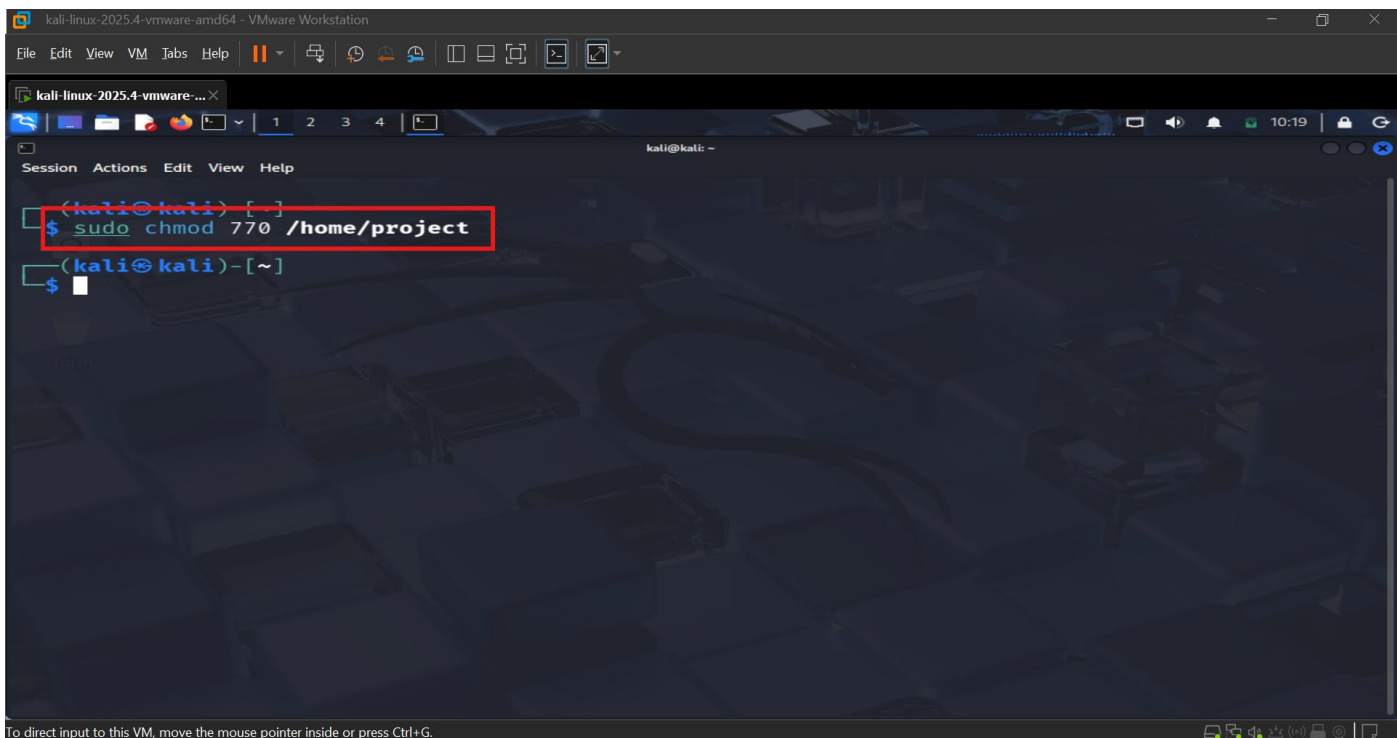
2.4.1 Assign group ownership to any one of the groups to the directory

Command: **sudo chown :proja /home/project**
sudo chown :projb /home/project



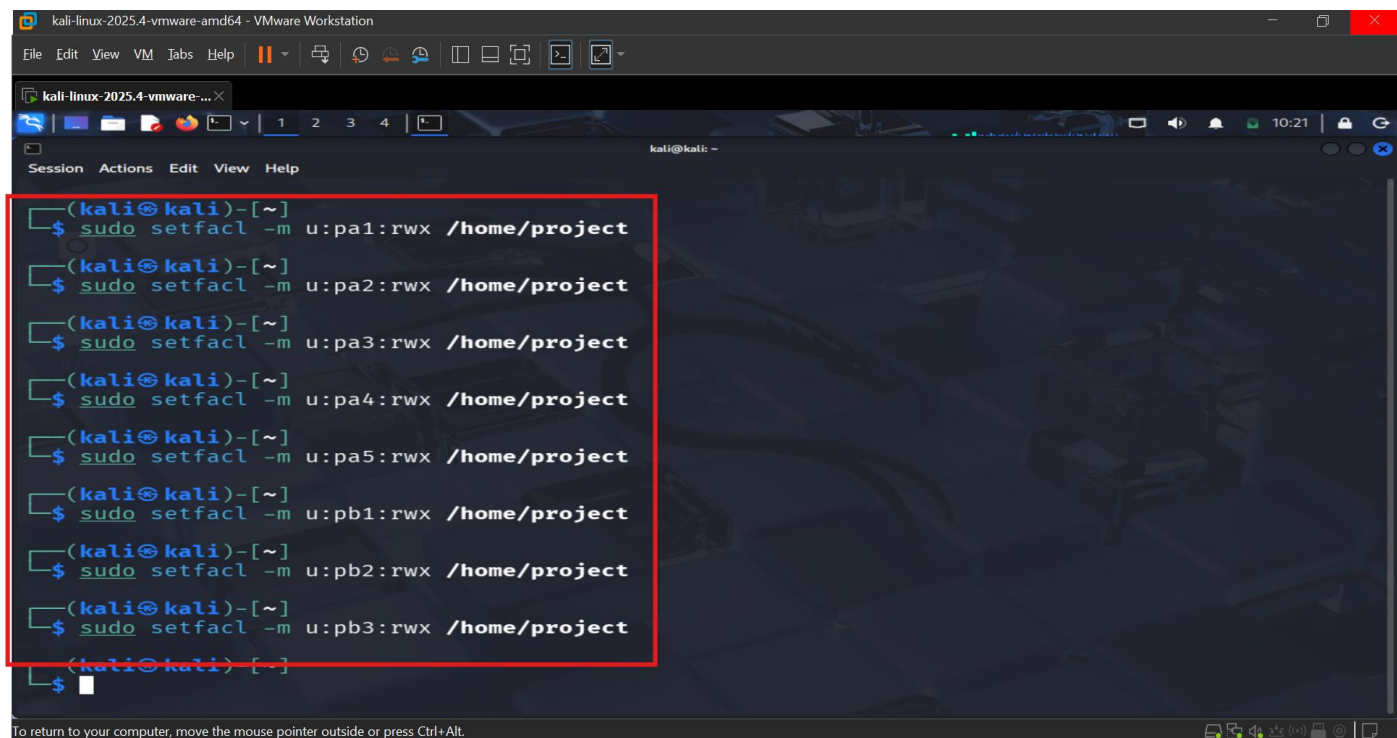
2.4.2 Set Directory Permissions

Command: **sudo chmod 770 /home/project**



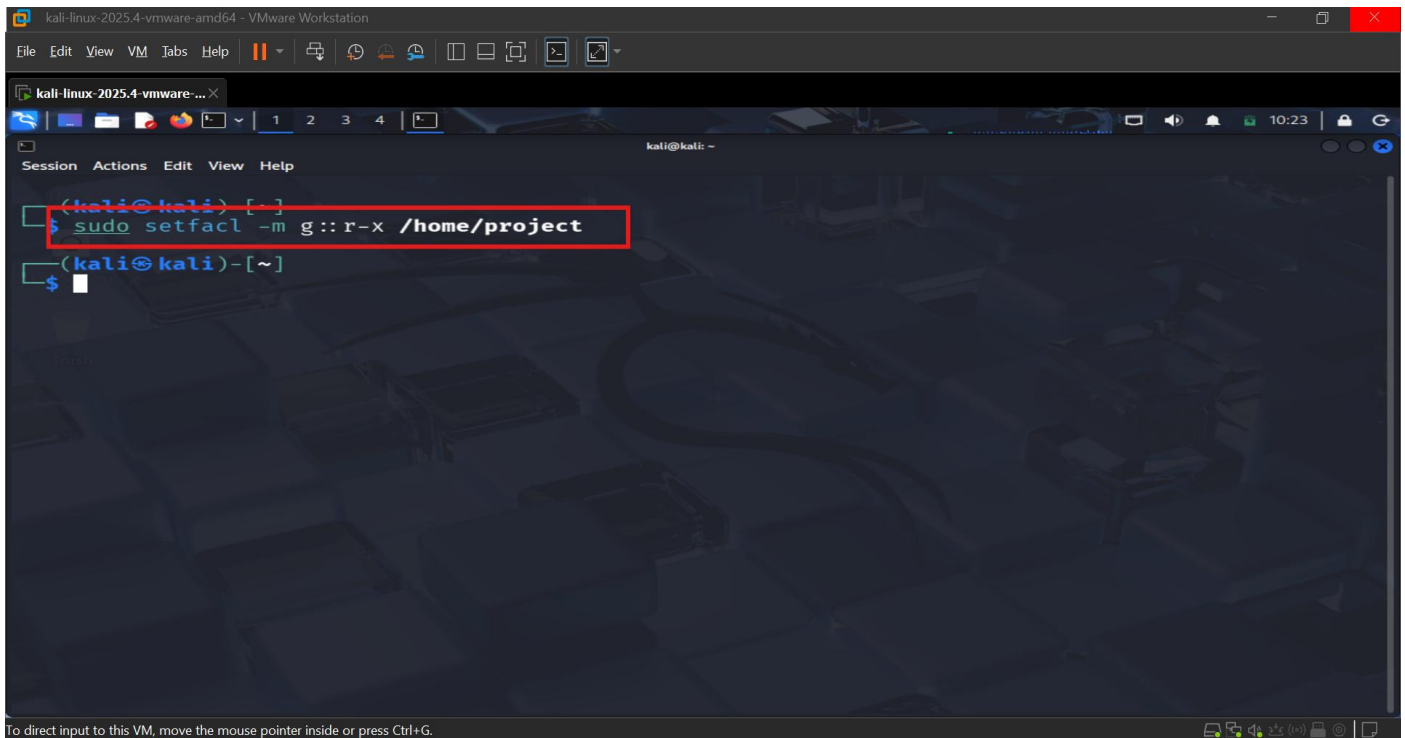
2.5 Configure ACLs to Restrict File Modifications

Command: `sudo setfacl -m u:pa1:rwX /home/project`
`sudo setfacl -m u:pa2:rwX /home/project`
`sudo setfacl -m u:pa3:rwX /home/project`
`sudo setfacl -m u:pa4:rwX /home/project`
`sudo setfacl -m u:pa5:rwX /home/project`
`sudo setfacl -m u:pb1:rwX /home/project`
`sudo setfacl -m u:pb2:rwX /home/project`
`sudo setfacl -m u:pb3:rwX /home/project`



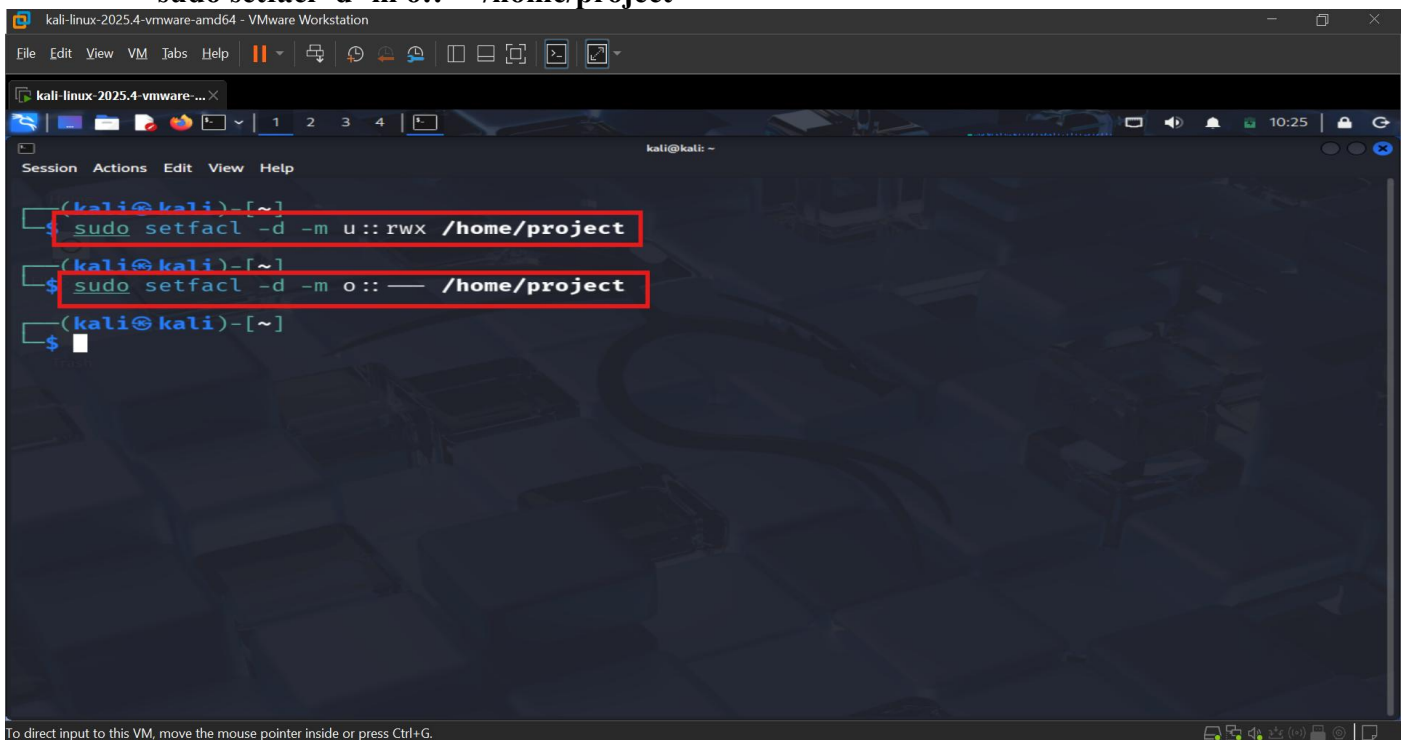
2.6 Restrict others to delete or modify the file

Command: `sudo setfacl -m g::r-x /home/project`



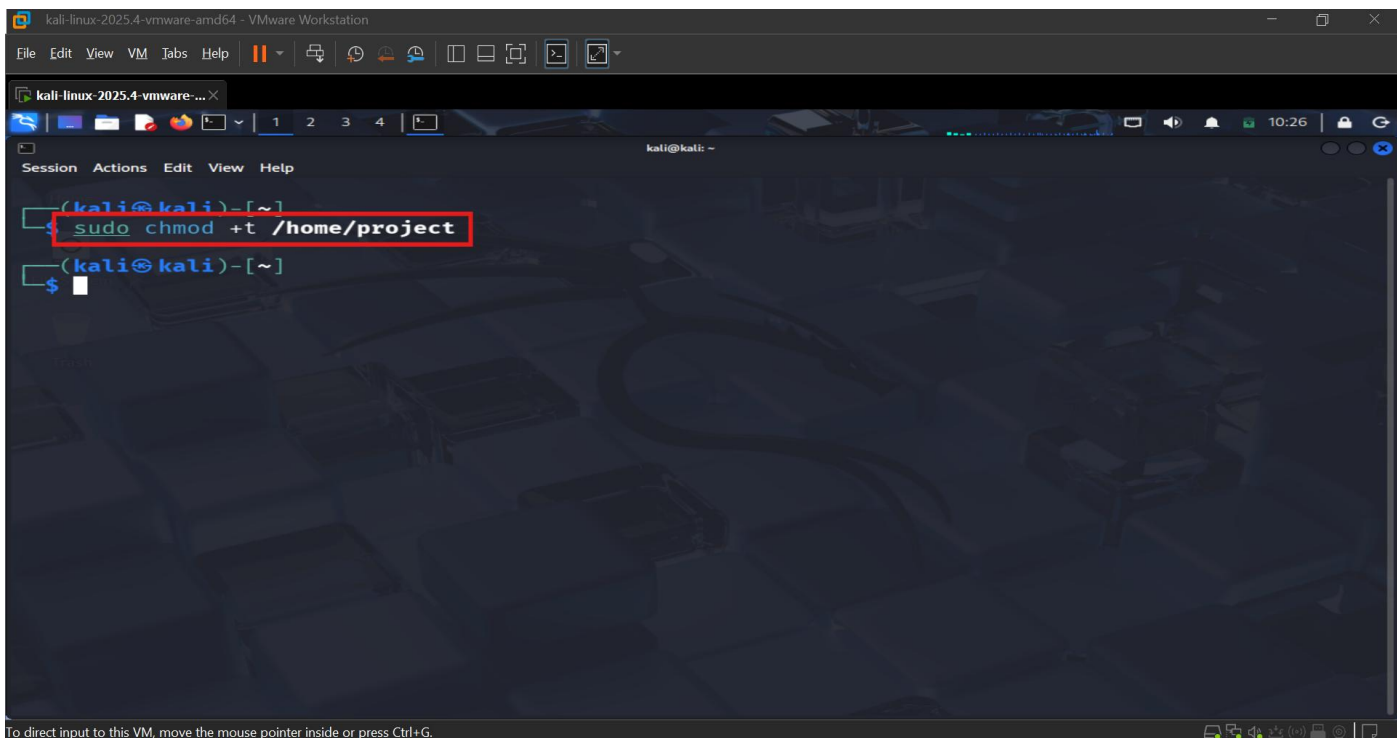
2.7 Apply Default ACLs

Command: **`sudo setfacl -d -m u::rwx /home/project`**
`sudo setfacl -d -m o::--- /home/project`



2.8 Set the Sticky Bit

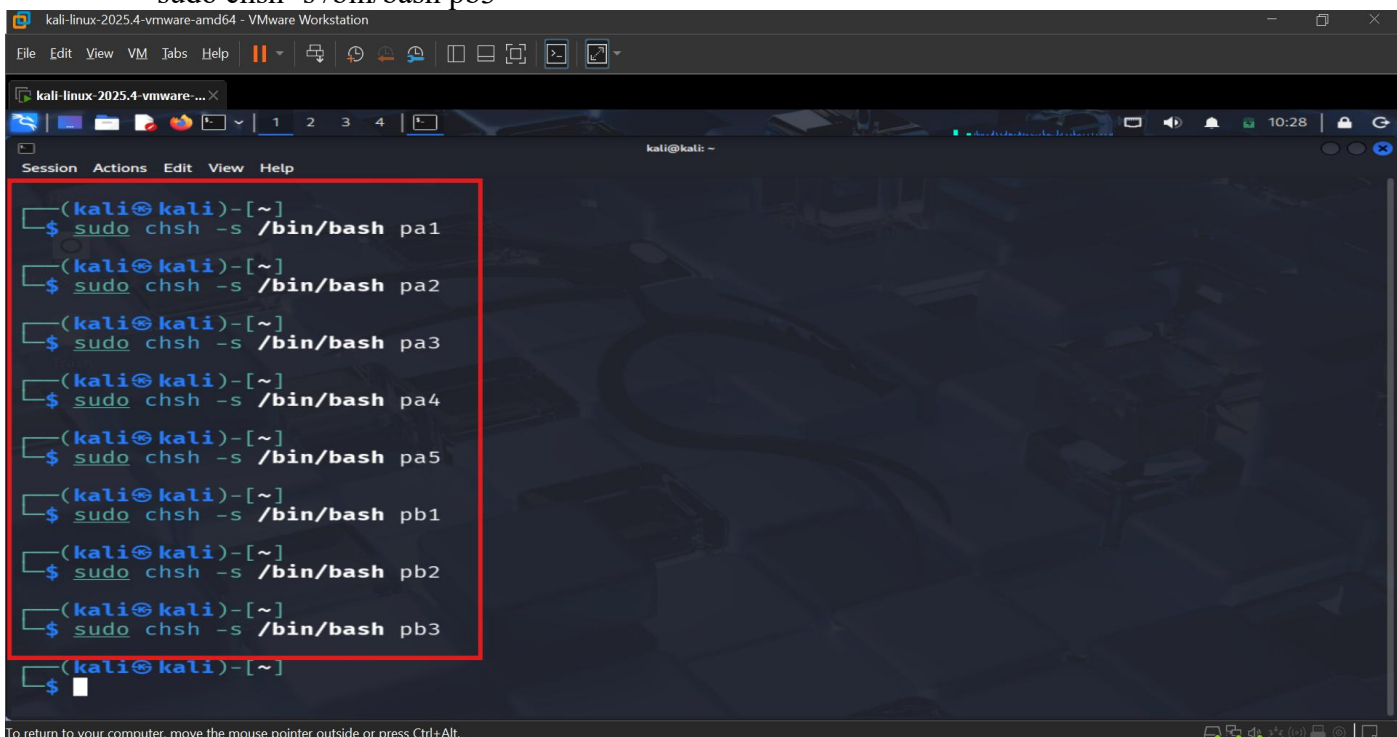
Command: **`sudo chmod +t /home/project`**



Step 3: Configuring Command History Settings

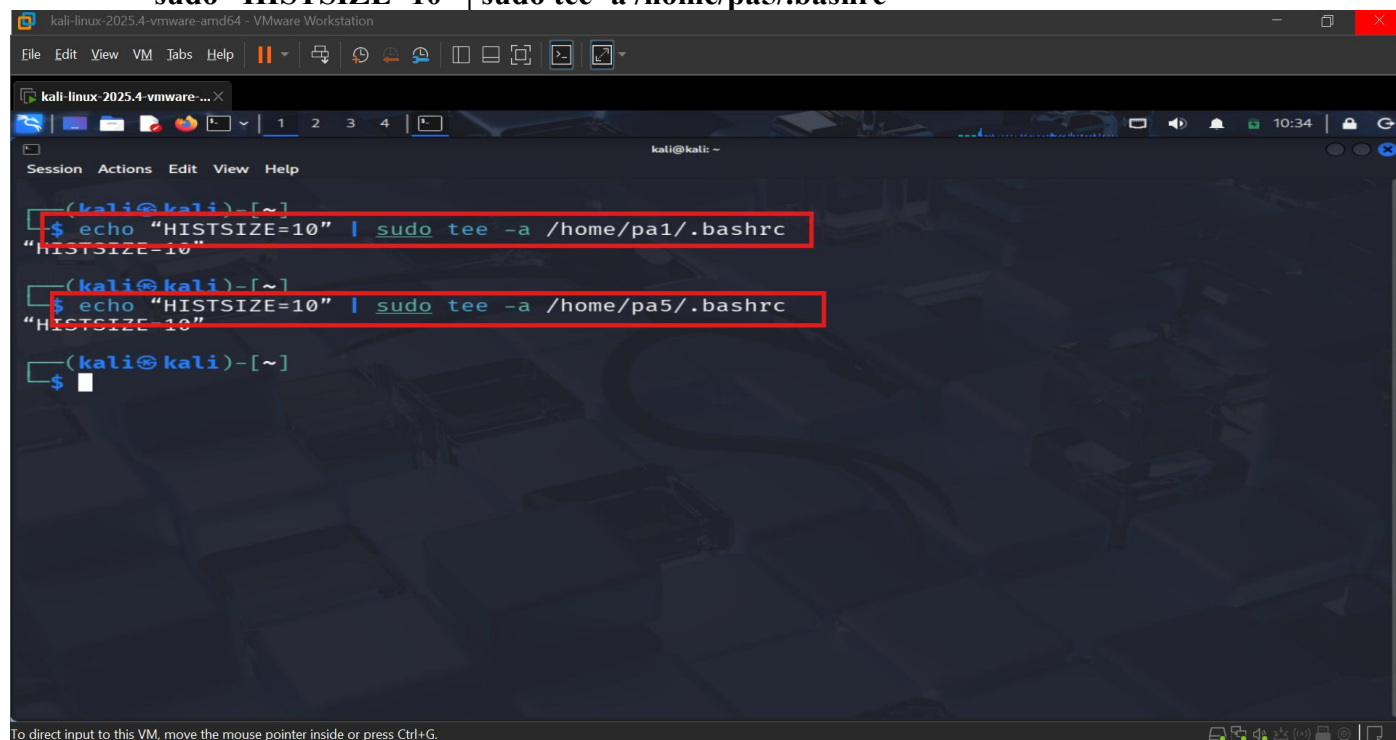
3.1 change the default shell of the user to /bin/bash

Command: sudo chsh -s /bin/bash pa1
sudo chsh -s /bin/bash pa2
sudo chsh -s /bin/bash pa3
sudo chsh -s /bin/bash pa4
sudo chsh -s /bin/bash pa5
sudo chsh -s /bin/bash pb1
sudo chsh -s /bin/bash pb2
sudo chsh -s /bin/bash pb3

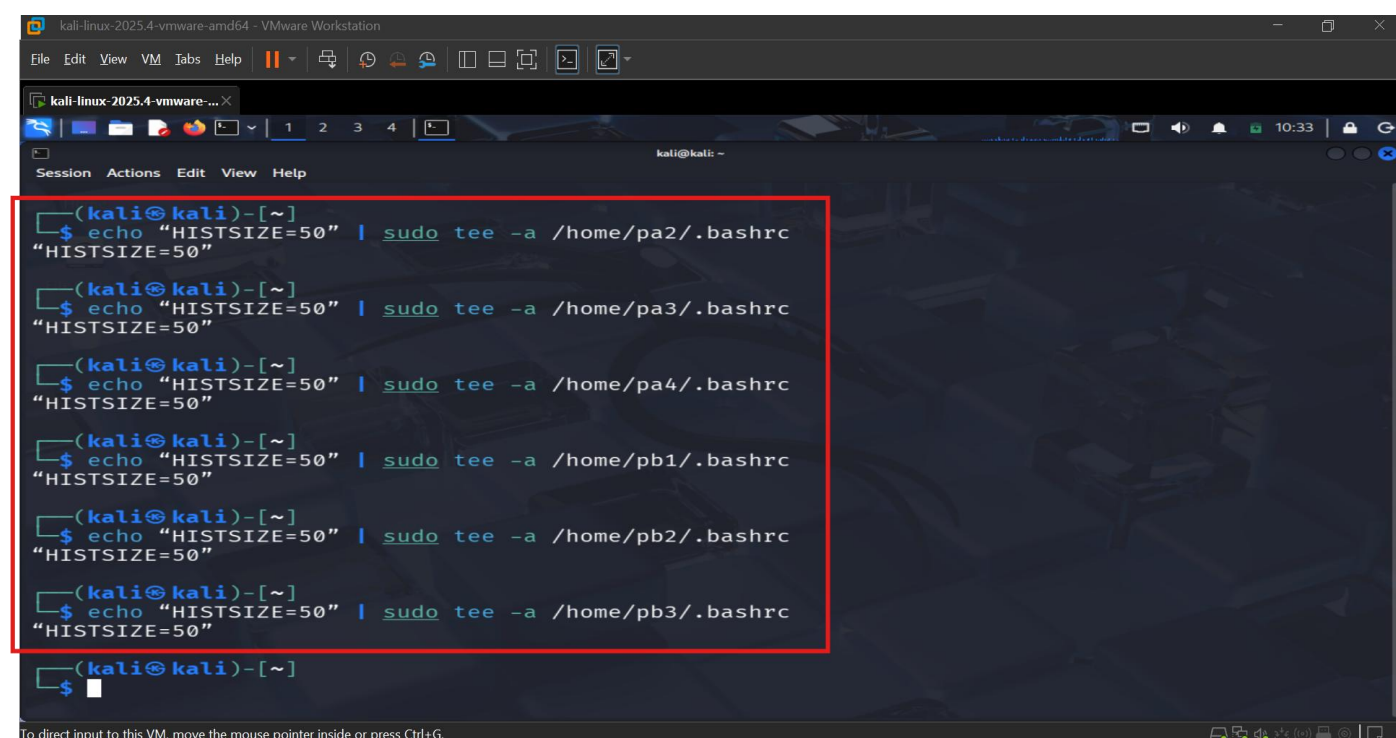


3.2 Set Command History Limits for Users(for pa1 and pa5)

Command: **sudo echo "HISTSIZE=10" | sudo tee -a /home/pa1/.bashrc**
sudo echo "HISTSIZE=10" | sudo tee -a /home/pa5/.bashrc



```
kali@kali: ~  
$ sudo echo "HISTSIZE=10" | sudo tee -a /home/pa1/.bashrc  
"HISTSIZE=10"  
$ sudo echo "HISTSIZE=10" | sudo tee -a /home/pa5/.bashrc  
"HISTSIZE=10"  
$
```

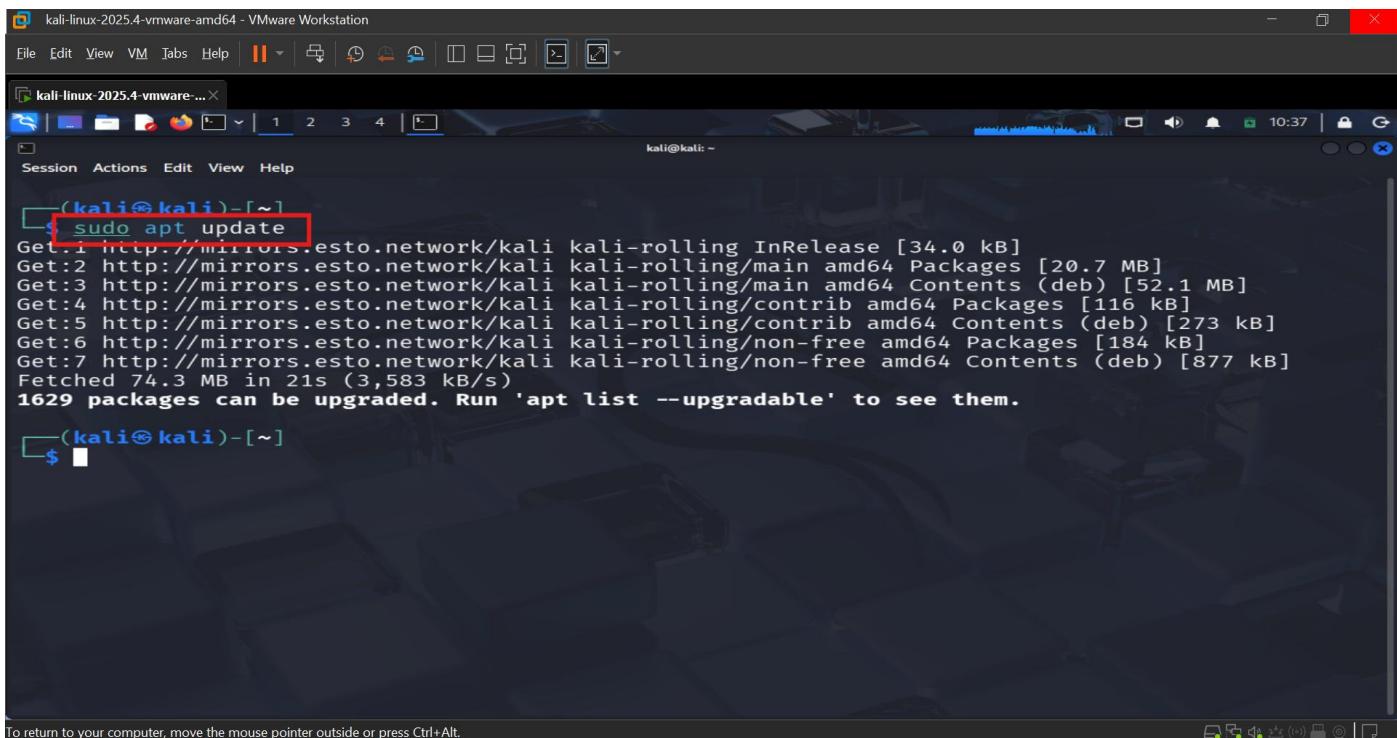


```
kali@kali: ~  
$ sudo echo "HISTSIZE=50" | sudo tee -a /home/pa2/.bashrc  
"HISTSIZE=50"  
$ sudo echo "HISTSIZE=50" | sudo tee -a /home/pa3/.bashrc  
"HISTSIZE=50"  
$ sudo echo "HISTSIZE=50" | sudo tee -a /home/pa4/.bashrc  
"HISTSIZE=50"  
$ sudo echo "HISTSIZE=50" | sudo tee -a /home/pb1/.bashrc  
"HISTSIZE=50"  
$ sudo echo "HISTSIZE=50" | sudo tee -a /home/pb2/.bashrc  
"HISTSIZE=50"  
$ sudo echo "HISTSIZE=50" | sudo tee -a /home/pb3/.bashrc  
"HISTSIZE=50"  
$
```

Step 4: Logging Security Violations

4.1 Update the os

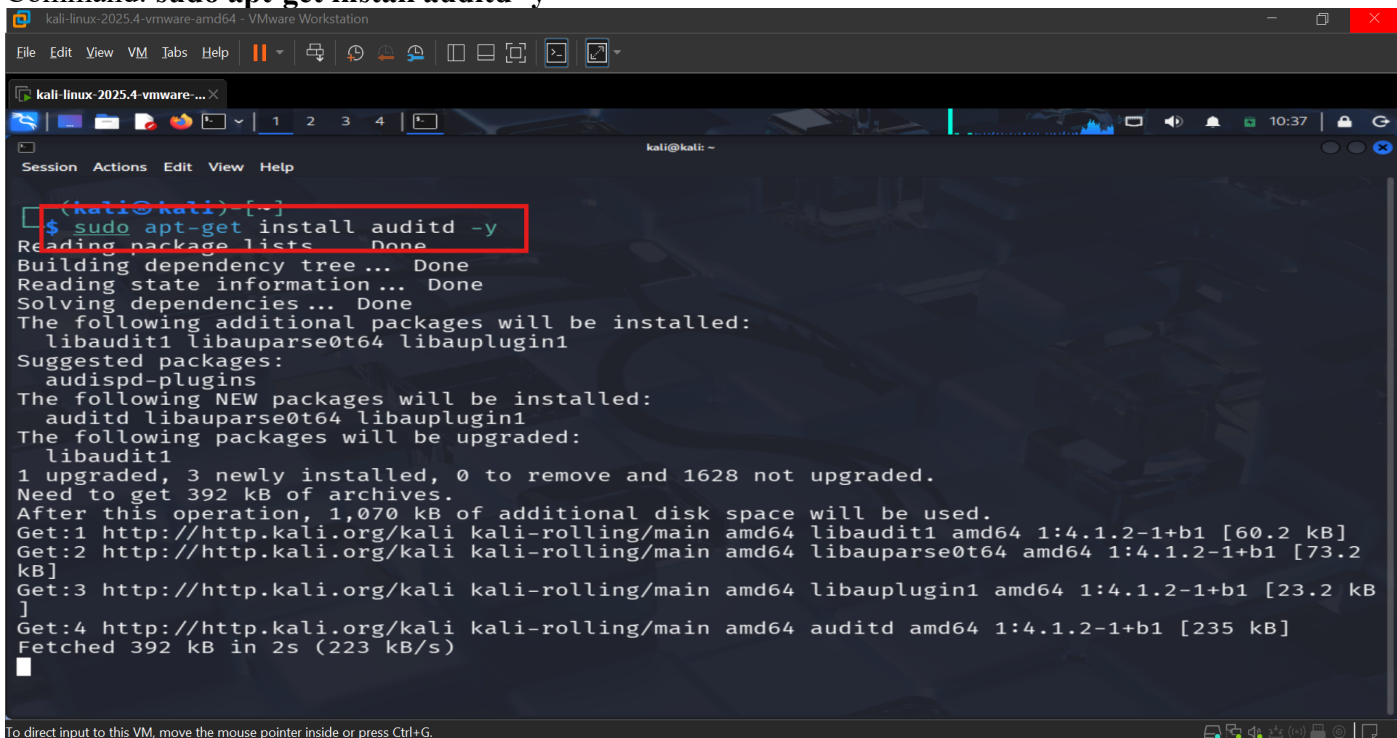
Command: **sudo apt update**



```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-...
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo apt update
Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [20.7 MB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]
Get:4 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents (deb) [273 kB]
Get:6 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [184 kB]
Get:7 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents (deb) [877 kB]
Fetched 74.3 MB in 21s (3,583 kB/s)
1629 packages can be upgraded. Run 'apt list --upgradable' to see them.
(kali㉿kali)-[~]
$
```

4.2 Install and Configure Auditd

Command: **sudo apt-get install auditd -y**

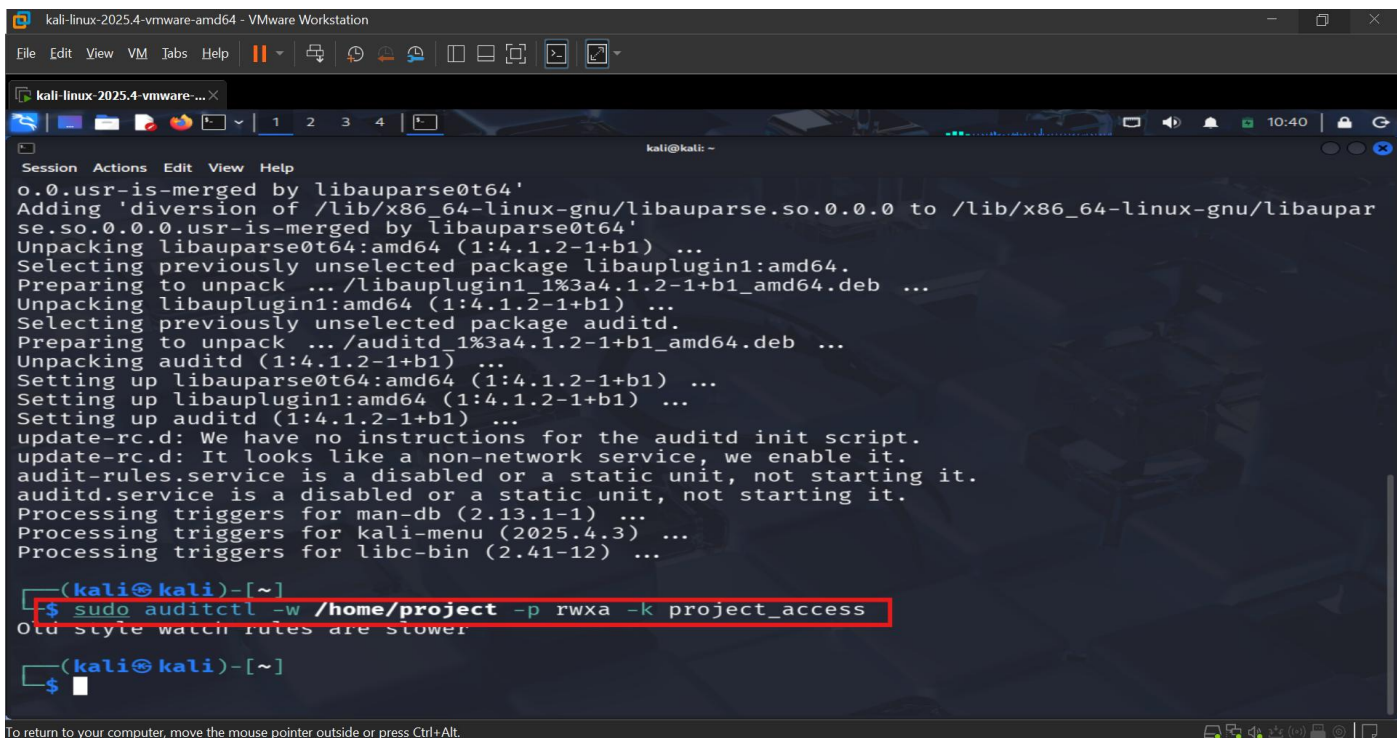


```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-...
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo apt-get install auditd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  libaudit1 libauparse0t64 libauplugin1
Suggested packages:
  audispd-plugins
The following NEW packages will be installed:
  auditd libauparse0t64 libauplugin1
The following packages will be upgraded:
  libaudit1
1 upgraded, 3 newly installed, 0 to remove and 1628 not upgraded.
Need to get 392 kB of archives.
After this operation, 1,070 kB of additional disk space will be used.
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libaudit1 amd64 1:4.1.2-1+b1 [60.2 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libauparse0t64 amd64 1:4.1.2-1+b1 [73.2 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libauplugin1 amd64 1:4.1.2-1+b1 [23.2 kB]
Get:4 http://http.kali.org/kali kali-rolling/main amd64 auditd amd64 1:4.1.2-1+b1 [235 kB]
Fetched 392 kB in 2s (223 kB/s)

```

4.3 Create Audit Rules to Log Access to the Project Directory

Command: **sudo auditctl -w /home/project -p rwx -k project_access**



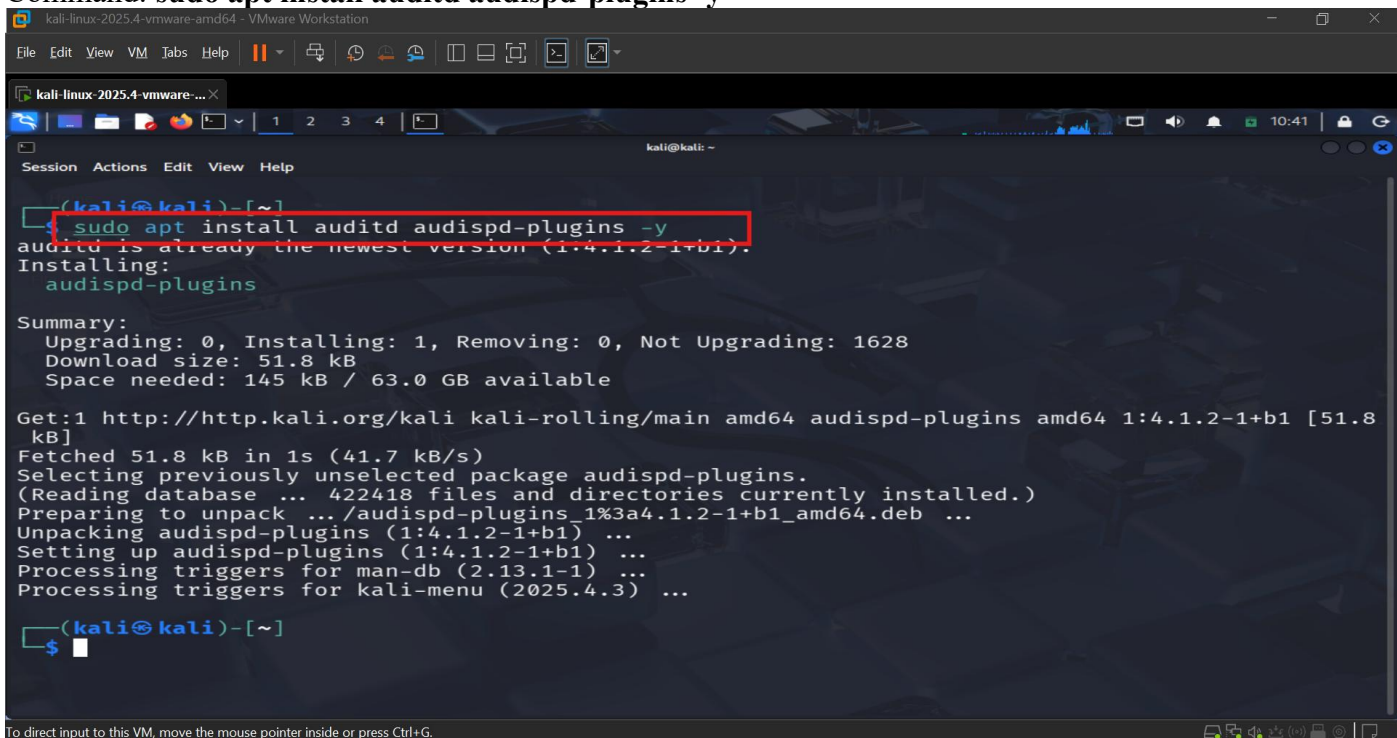
```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-...
Session Actions Edit View Help
kali@kali: ~
o.0.usr-is-merged by libauparse0t64'
Adding 'diversion of /lib/x86_64-linux-gnu/libauparse.so.0.0.0 to /lib/x86_64-linux-gnu/libaupar
se.so.0.0.0.usr-is-merged by libauparse0t64'
Unpacking libauparse0t64:amd64 (1:4.1.2-1+b1) ...
Selecting previously unselected package libauplugin1:amd64.
Preparing to unpack .../libauplugin1_1%3a4.1.2-1+b1_amd64.deb ...
Unpacking libauplugin1:amd64 (1:4.1.2-1+b1) ...
Selecting previously unselected package auditd.
Preparing to unpack .../auditd_1%3a4.1.2-1+b1_amd64.deb ...
Unpacking auditd (1:4.1.2-1+b1) ...
Setting up libauparse0t64:amd64 (1:4.1.2-1+b1) ...
Setting up libauplugin1:amd64 (1:4.1.2-1+b1) ...
Setting up auditd (1:4.1.2-1+b1) ...
update-rc.d: We have no instructions for the auditd init script.
update-rc.d: It looks like a non-network service, we enable it.
audit-rules.service is a disabled or a static unit, not starting it.
auditd.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for libc-bin (2.41-12) ...

(kali@kali)-[~]
$ sudo auditctl -w /home/project -p rwx -k project_access
Old style watch rules are slower

(kali@kali)-[~]
$
```

4.4 Install the Audit Daemon (auditd) dependencies.

Command: **sudo apt install auditd audispd-plugins -y**



```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-...
Session Actions Edit View Help
kali@kali: ~

(kali@kali)-[~]
$ sudo apt install auditd audispd-plugins -y
auditd is already the newest version (1:4.1.2-1+b1).
Installing:
  audispd-plugins

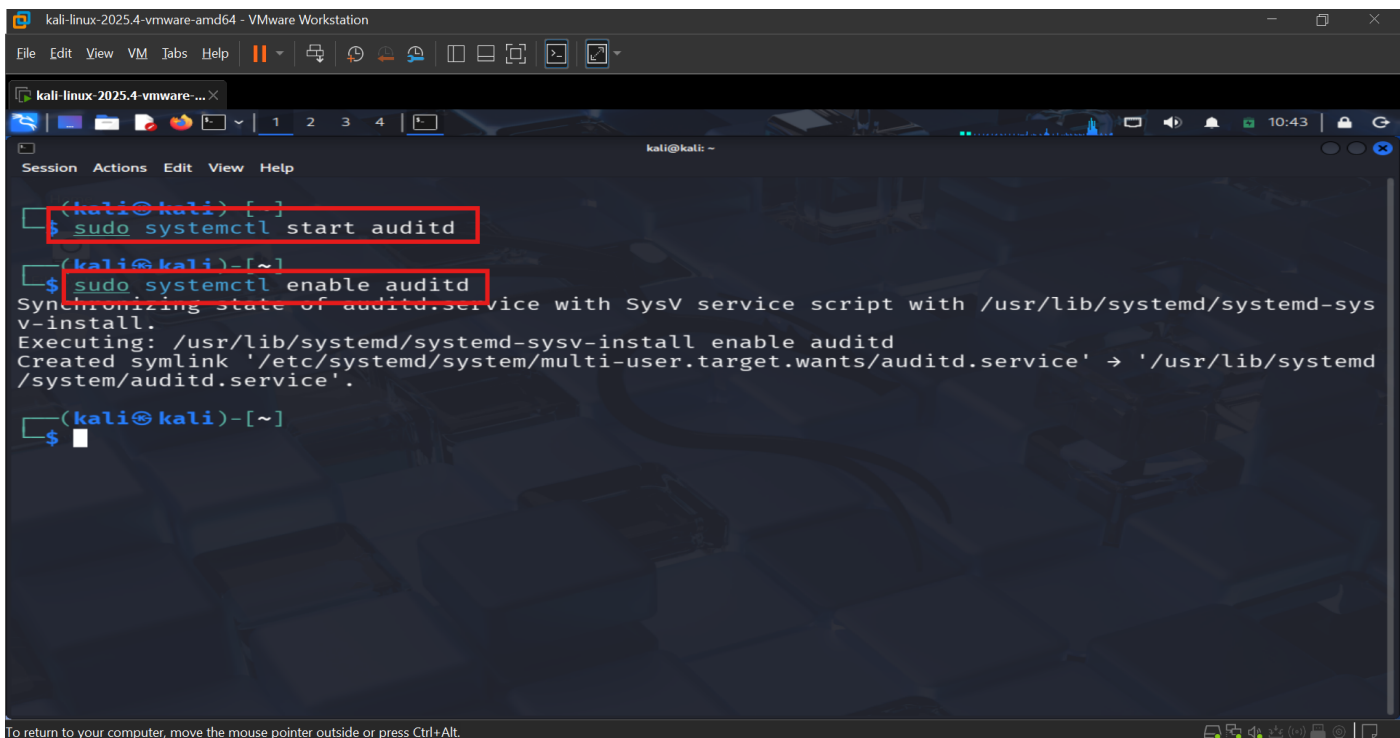
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1628
  Download size: 51.8 kB
  Space needed: 145 kB / 63.0 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 audispd-plugins amd64 1:4.1.2-1+b1 [51.8
kB]
Fetched 51.8 kB in 1s (41.7 kB/s)
Selecting previously unselected package audispd-plugins.
(Reading database ... 422418 files and directories currently installed.)
Preparing to unpack .../audispd-plugins_1%3a4.1.2-1+b1_amd64.deb ...
Unpacking audispd-plugins (1:4.1.2-1+b1) ...
Setting up audispd-plugins (1:4.1.2-1+b1) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

(kali@kali)-[~]
$
```

4.5 Start and Enable the auditd Service

Command: **sudo systemctl start auditd**
sudo systemctl enable auditd



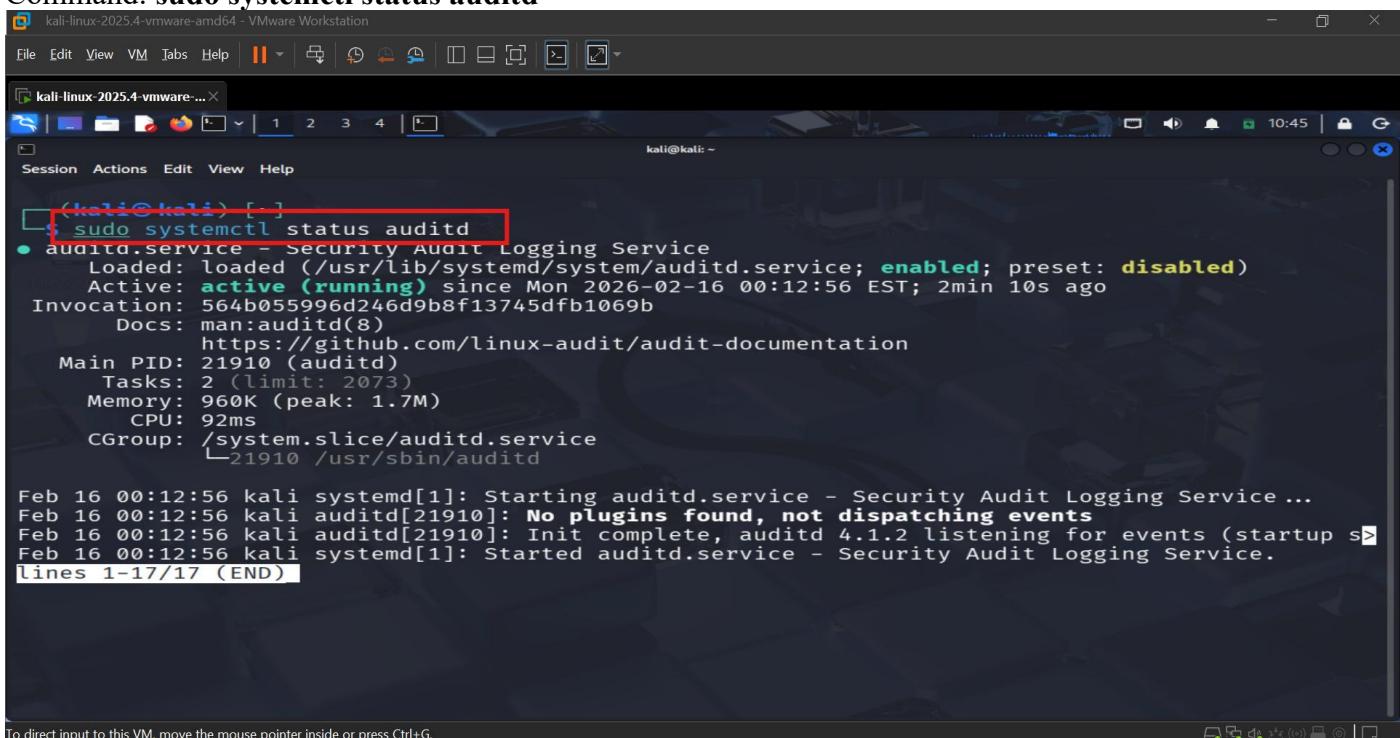
```
(kali@kali) [~]
$ sudo systemctl start auditd

(kali@kali) [~]
$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' -> '/usr/lib/systemd/system/auditd.service'.

(kali@kali) [~]
$
```

4.6 Verify auditd Service Status

Command: **sudo systemctl status auditd**

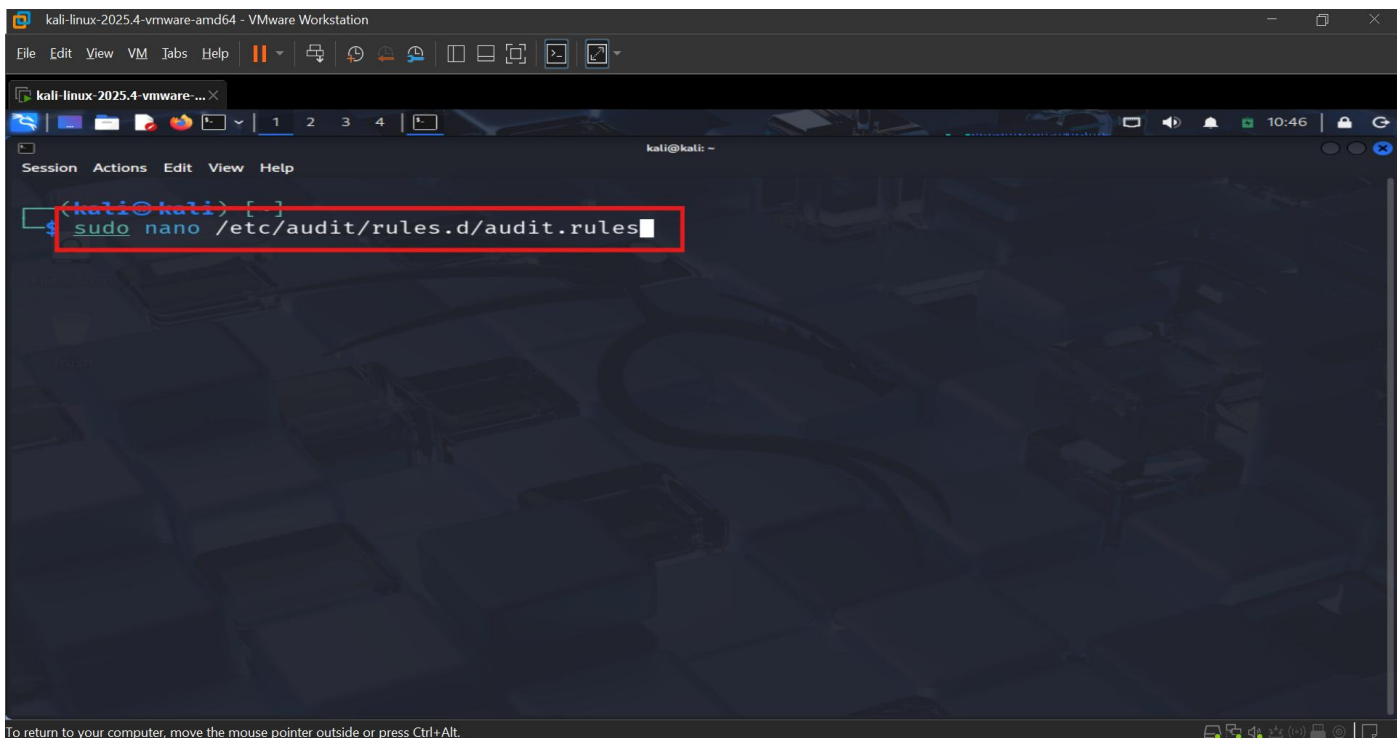


```
(kali@kali) [~]
$ sudo systemctl status auditd
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-02-16 00:12:56 EST; 2min 10s ago
     Invocation: 564b055996d246d9b8f13745dfb1069b
       Docs: man:auditd(8)
            https://github.com/linux-audit/audit-documentation
   Main PID: 21910 (auditd)
      Tasks: 2 (limit: 2073)
     Memory: 960K (peak: 1.7M)
        CPU: 92ms
    CGroup: /system.slice/auditd.service
            └─21910 /usr/sbin/auditd

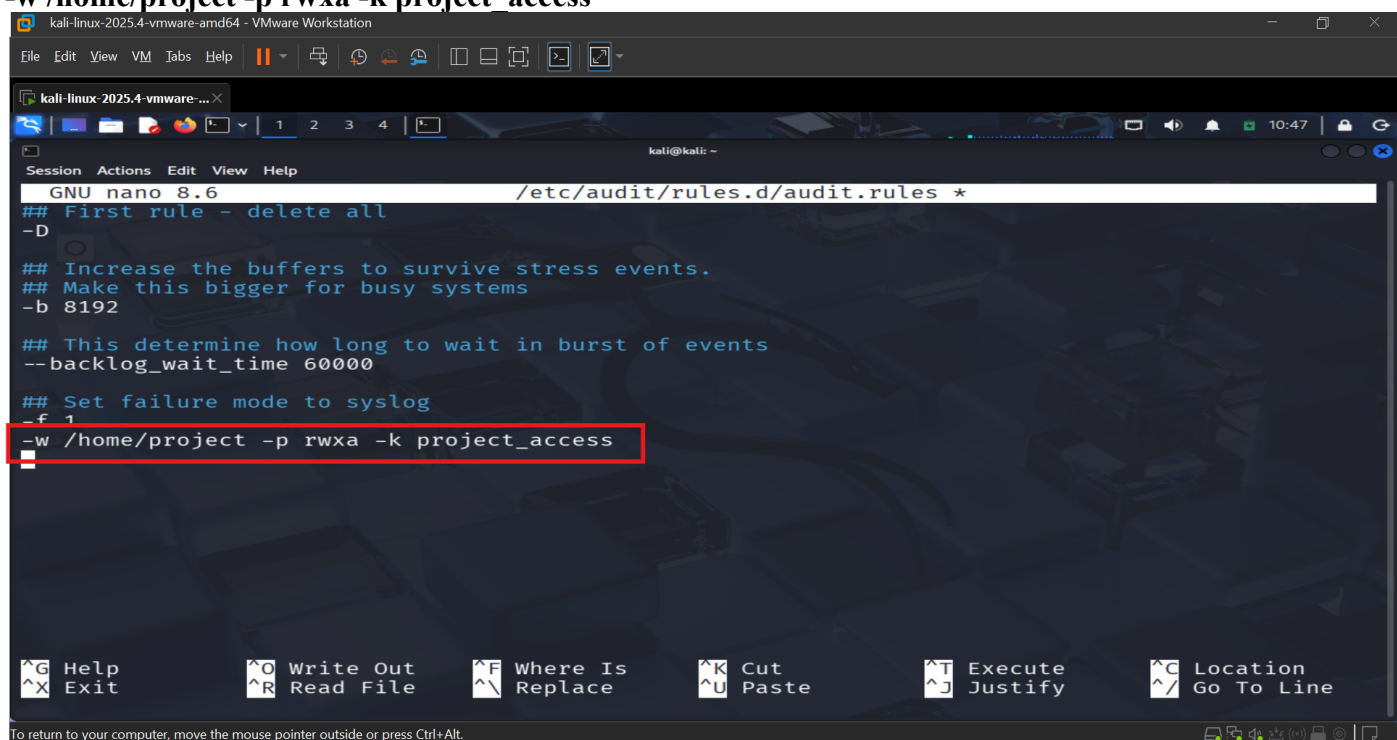
Feb 16 00:12:56 kali systemd[1]: Starting auditd.service - Security Audit Logging Service ...
Feb 16 00:12:56 kali auditd[21910]: No plugins found, not dispatching events
Feb 16 00:12:56 kali auditd[21910]: Init complete, auditd 4.1.2 listening for events (startup s
Feb 16 00:12:56 kali systemd[1]: Started auditd.service - Security Audit Logging Service.
lines 1-17/17 (END)
```

4.7 Create and Verify Audit Rules

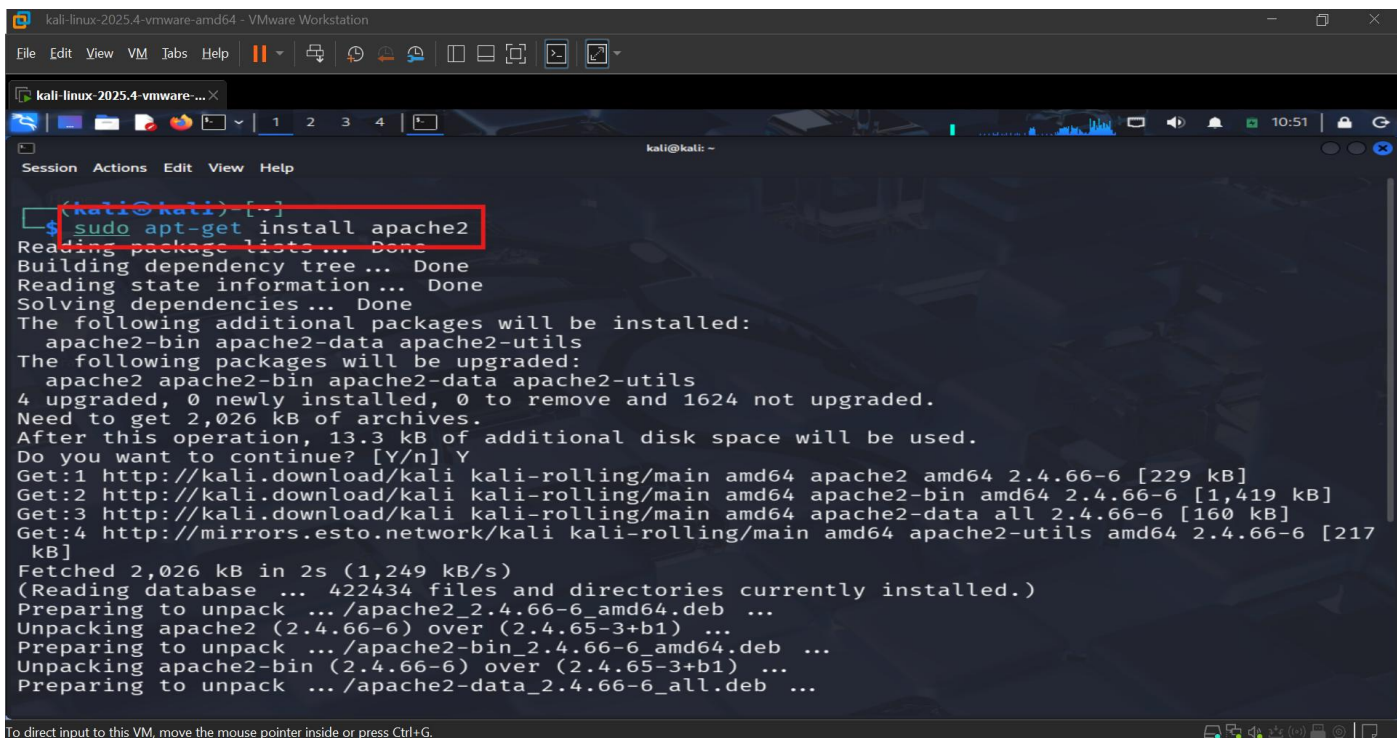
Command: **sudo nano /etc/audit/rules.d/audit.rules**



Add the following line at the end of the file
-w /home/project -p rwx -k project_access

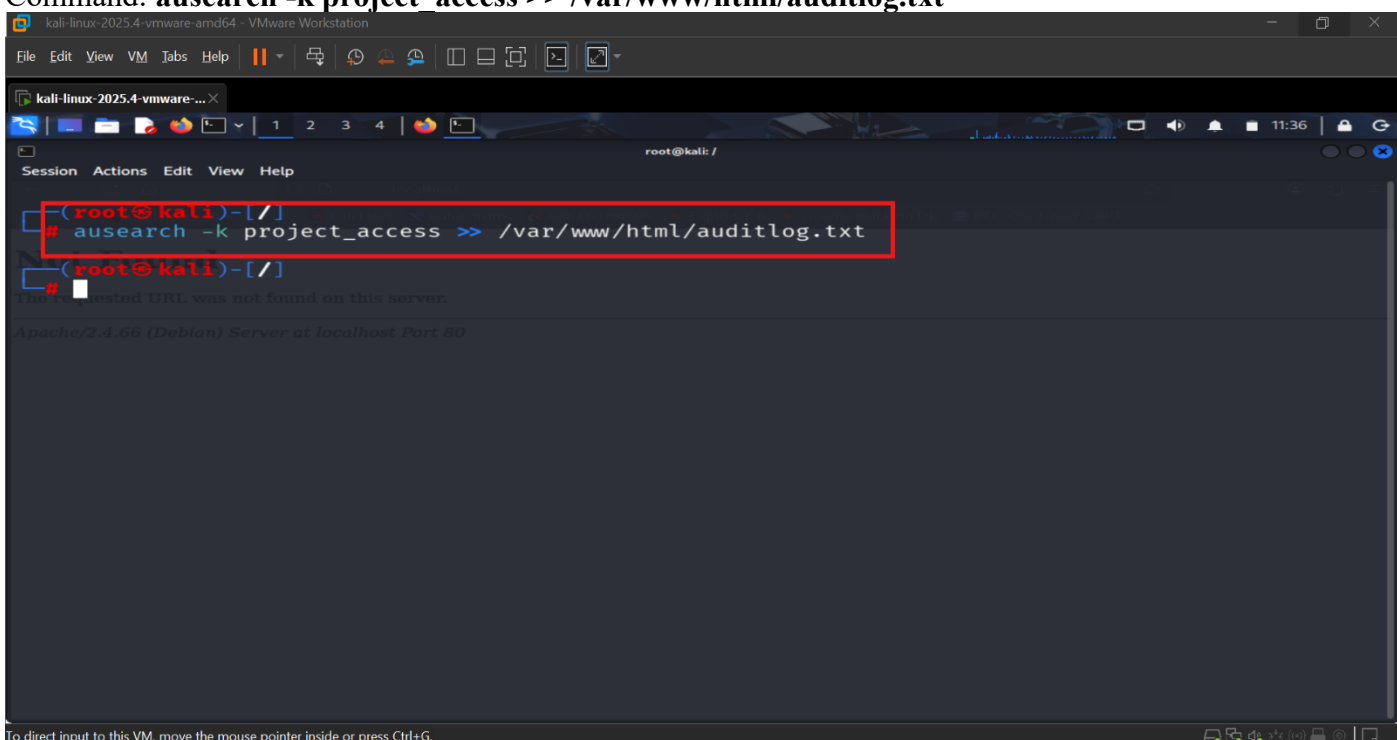


4.8 Restart the auditd service to apply the changes
Command: **sudo systemctl restart auditd**



```
kali@kali:~$ sudo apt-get install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils
The following packages will be upgraded:
  apache2 apache2-bin apache2-data apache2-utils
4 upgraded, 0 newly installed, 0 to remove and 1624 not upgraded.
Need to get 2,026 kB of archives.
After this operation, 13.3 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://kali.download/kali kali-rolling/main amd64 apache2 amd64 2.4.66-6 [229 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 apache2-bin amd64 2.4.66-6 [1,419 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 apache2-data all 2.4.66-6 [160 kB]
Get:4 http://mirrors.esto.network/kali kali-rolling/main amd64 apache2-utils amd64 2.4.66-6 [217 kB]
Fetched 2,026 kB in 2s (1,249 kB/s)
(Reading database ... 422434 files and directories currently installed.)
Preparing to unpack .../apache2_2.4.66-6_amd64.deb ...
Unpacking apache2 (2.4.66-6) over (2.4.65-3+b1) ...
Preparing to unpack .../apache2-bin_2.4.66-6_amd64.deb ...
Unpacking apache2-bin (2.4.66-6) over (2.4.65-3+b1) ...
Preparing to unpack .../apache2-data_2.4.66-6_all.deb ...
```

5.2 Run ausearch Command and transfer the logs to a
Command: **ausearch -k project access >> /var/www/html/auditlog.txt**



```
root@kali: /
# ausearch -k project_access >> /var/www/html/auditlog.txt
root@kali: /
The requested URL was not found on this server.
Apache/2.4.66 (Debian) Server at localhost Port 80
```

5.3 Using Crontab for Automation
Command: **crontab -e**

```
(root@kali)-[/]  
# ausearch -k project_access >> /var/www/html/auditlog.txt  
(root@kali)-[/]  
# crontab -e
```

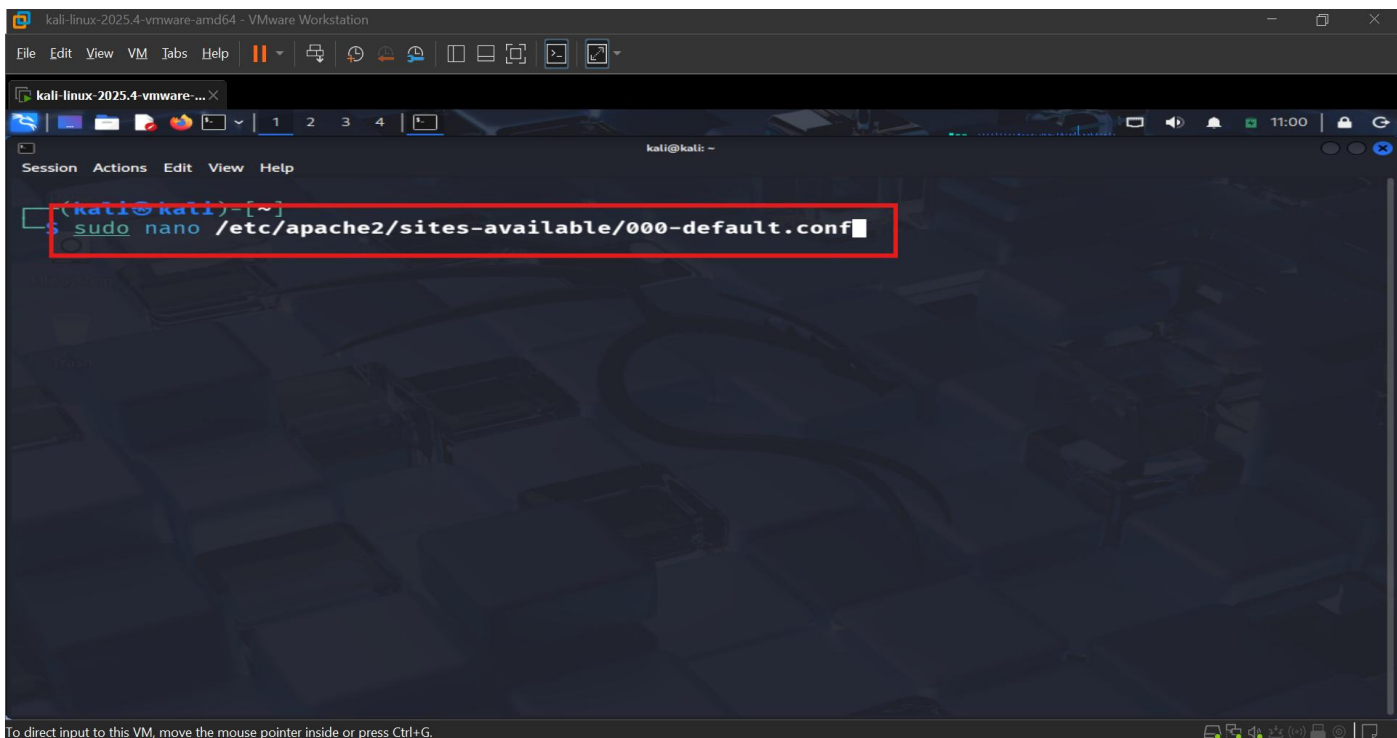
Within the crontab file, add this line

`*/5 * * * * ausearch -k project_access >> /var/www/html/auditlog.txt`

```
GNU nano 8.6 /tmp/crontab.b0MvXl/crontab *  
# Each task to run has to be defined through a single line  
# indicating with different fields when the task will be run  
# and what command to run for the task  
#  
# To define the time you can provide concrete values for  
# minute (m), hour (h), day of month (dom), month (mon),  
# and day of week (dow) or use '*' in these fields (for 'any').  
# Notice that tasks will be started based on the cron's system  
# daemon's notion of time and timezones.  
#  
# Output of the crontab jobs (including errors) is sent through  
# email to the user the crontab file belongs to (unless redirected).  
#  
# For example, you can run a backup of all your user accounts  
# at 5 a.m every week with:  
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/  
#  
# For more information see the manual pages of crontab(5) and cron(8)  
#  
# m h dom mon dow command  
*/5 * * * * ausearch -k project_access >> /var/www/html/auditlog.txt
```

5.4 Edit the Apache configuration file

Command: **`sudo nano /etc/apache2/sites-available/000-default.conf`**



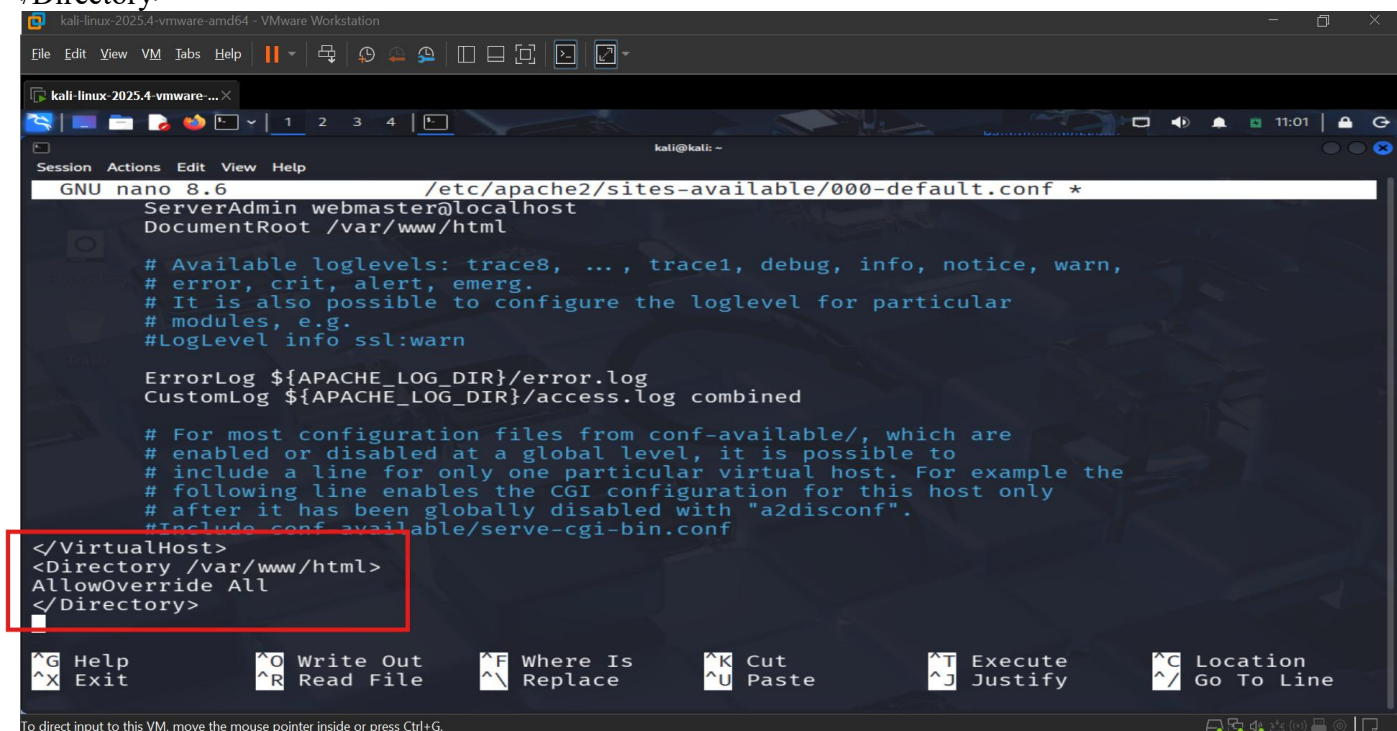
Add or modify the following lines:

</VirtualHost>

<Directory /var/www/html>

AllowOverride All

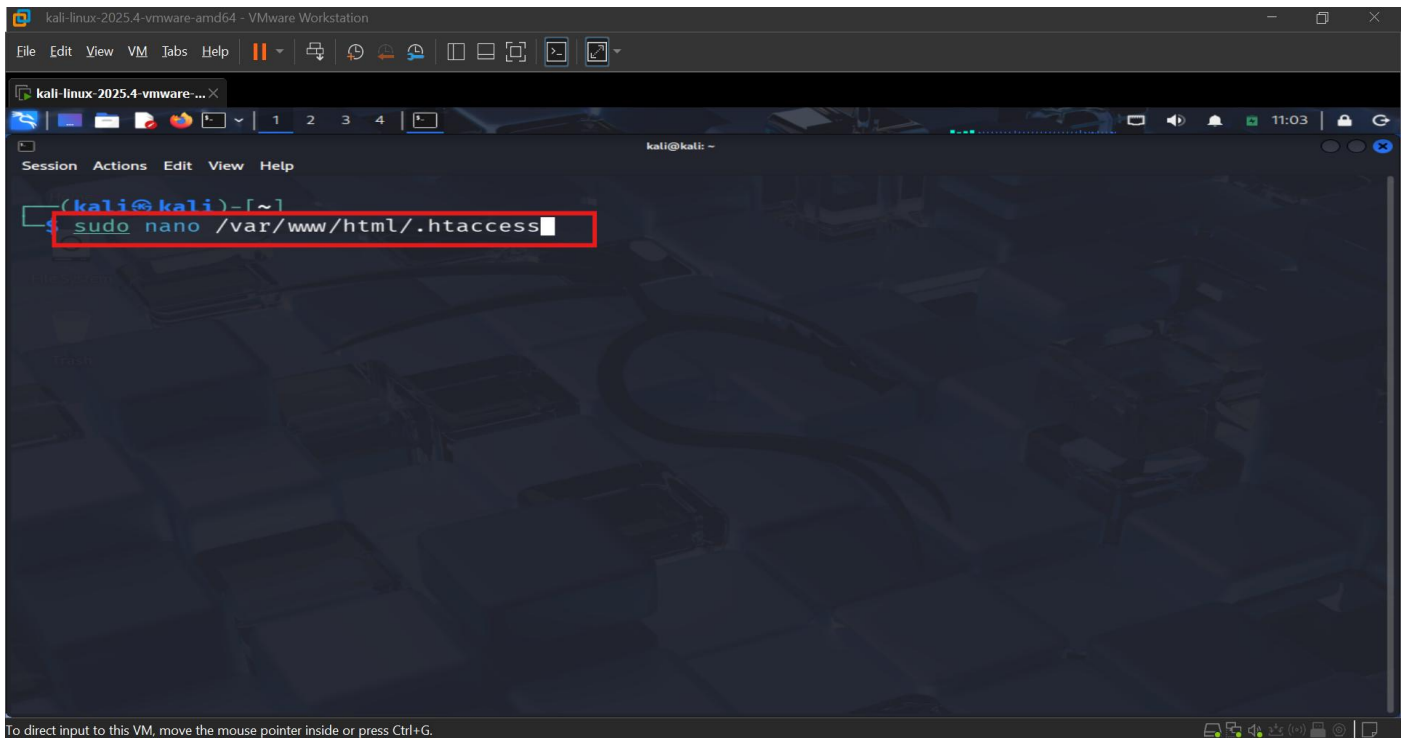
</Directory>



5.5 Secure Access by editing .htaccess

Create .htaccess file in /var/www/html:

Command: **sudo nano /var/www/html/.htaccess**



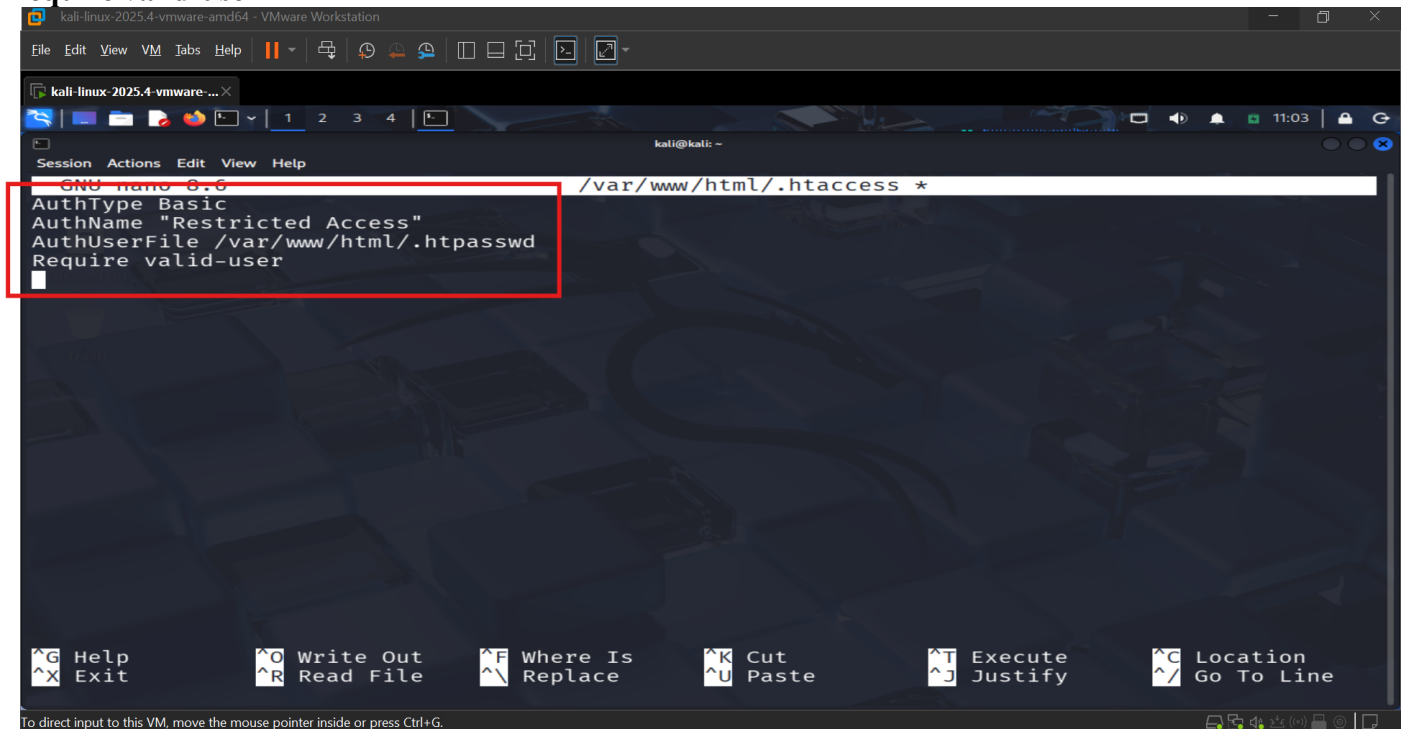
Add or modify the following lines:

AuthType Basic

AuthName "Restricted Access"

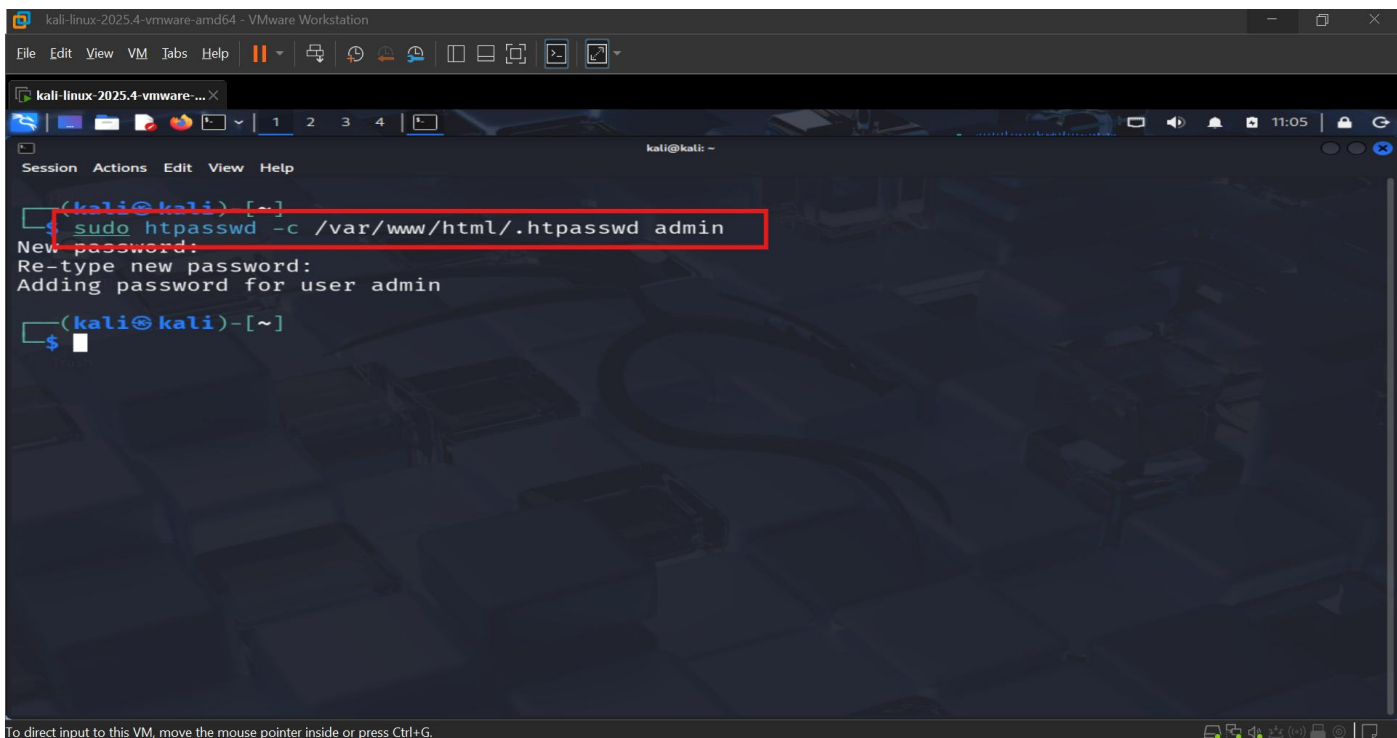
AuthUserFile /var/www/html/.htpasswd

Require valid-user



5.6 Create .htpasswd File for Authentication

Command: **sudo htpasswd -c /var/www/html/.htpasswd admin**

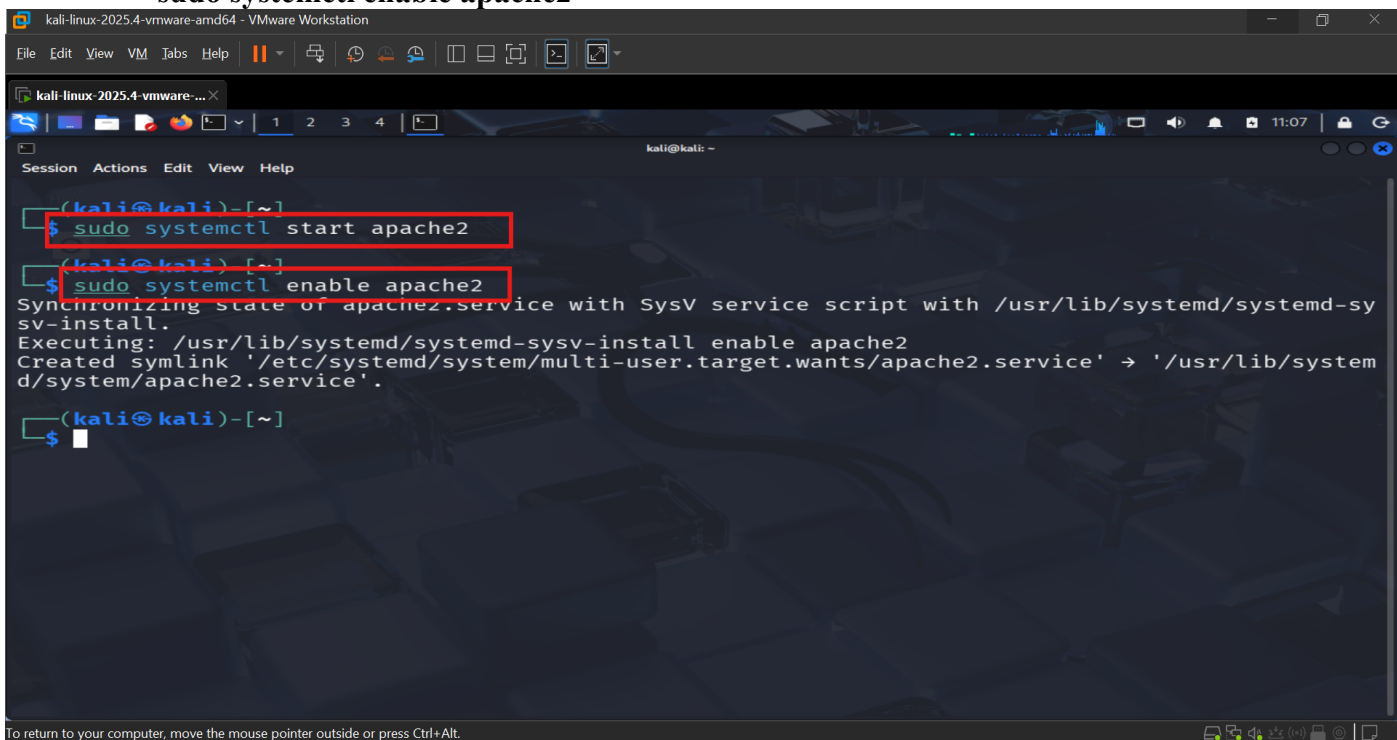


```
(kali@kali)-[~]  
$ sudo htpasswd -c /var/www/html/.htpasswd admin  
New password:  
Re-type new password:  
Adding password for user admin  
(kali@kali)-[~]  
$
```

5.7 Start and enable apache2

Command: **sudo systemctl start apache2**

sudo systemctl enable apache2



```
(kali@kali)-[~]  
$ sudo systemctl start apache2  
(kali@kali)-[~]  
$ sudo systemctl enable apache2  
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2  
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' -> '/usr/lib/systemd/system/apache2.service'.  
(kali@kali)-[~]  
$
```

Step 6: Testing and Validation

To confirm that your configuration is properly set up, you should conduct several tests to validate that permissions, command history settings, and logging are functioning as expected.

6.1: Test File Creation and Modification:

Log in as each user (pa1, pa2, pa3, pa4, pa5, pb1, pb2, pb3) and try creating a file in the /home/project directory.

Example:

- Log in as a **pa1** user using the command: **su - pa1**

- Create a text file using the command: **touch /home/project/testfile.txt**

```

kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-...
pa1@kali: ~
(kali@kali)-[~]
$ su - pa1
Password:
"HISTSIZE=10": command not found
"HISTSIZE=10": command not found
(pa1@kali)-[~]
$ touch /home/project/testfile.txt
(pa1@kali)-[~]
$

```

Now try to remove the created file through another user.

- First login as any user using the command: **su - pa2**
- Remove the text file using the command: **rm -f /home/project/testfile.txt**

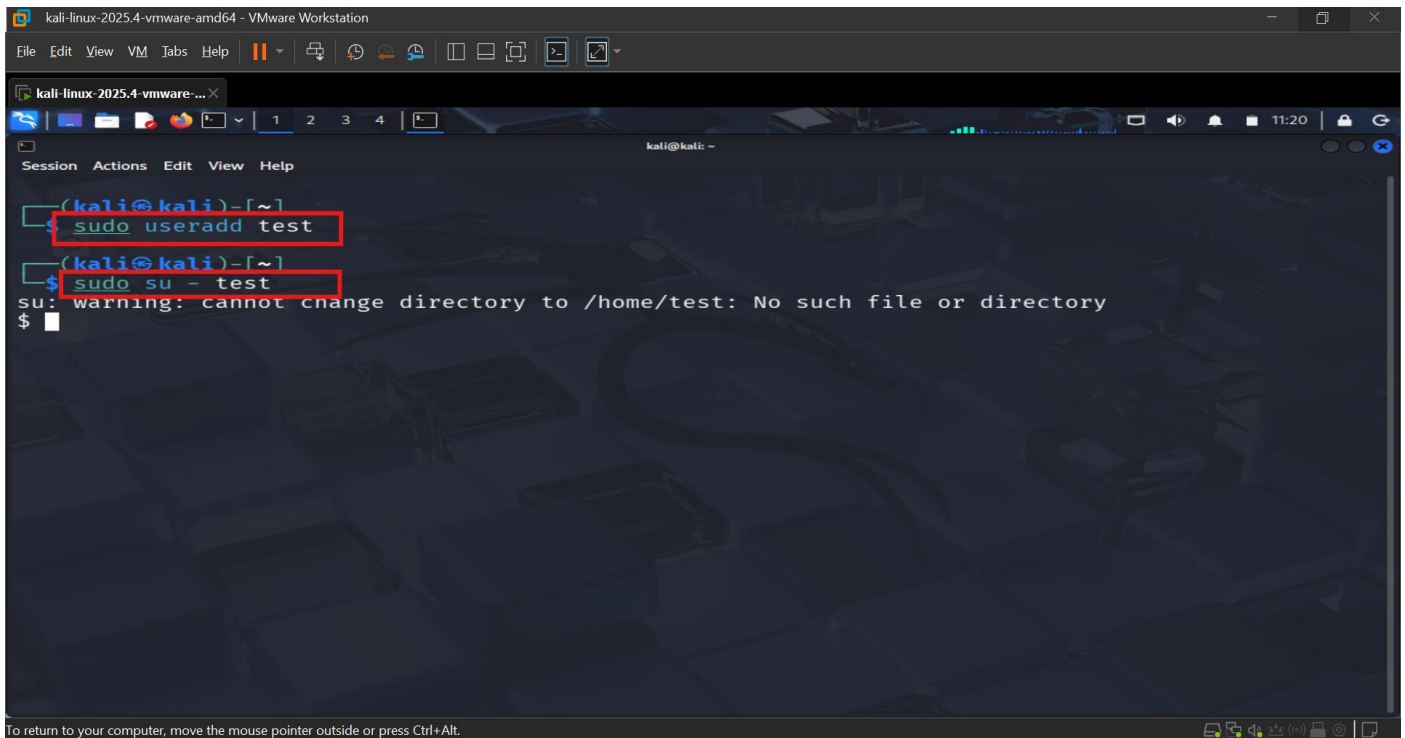
```

kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.4-vmware-...
pb1@kali: ~
(kali@kali)-[~]
$ su - pa2
Password:
"HISTSIZE=50": command not found
(pa2@kali)-[~]
$ rm /home/project/testfile.txt
rm: remove write-protected regular empty file '/home/project/testfile.txt'?
(pa2@kali)-[~]
$ rm -f /home/project/testfile.txt
rm: cannot remove '/home/project/testfile.txt': Operation not permitted
(pa2@kali)-[~]
$ exit
logout
(kali@kali)-[~]
$ su - pb1
Password:
"HISTSIZE=50": command not found
(pb1@kali)-[~]
$ rm -f /home/project/testfile.txt
rm: cannot remove '/home/project/testfile.txt': Operation not permitted
(pb1@kali)-[~]
$

```

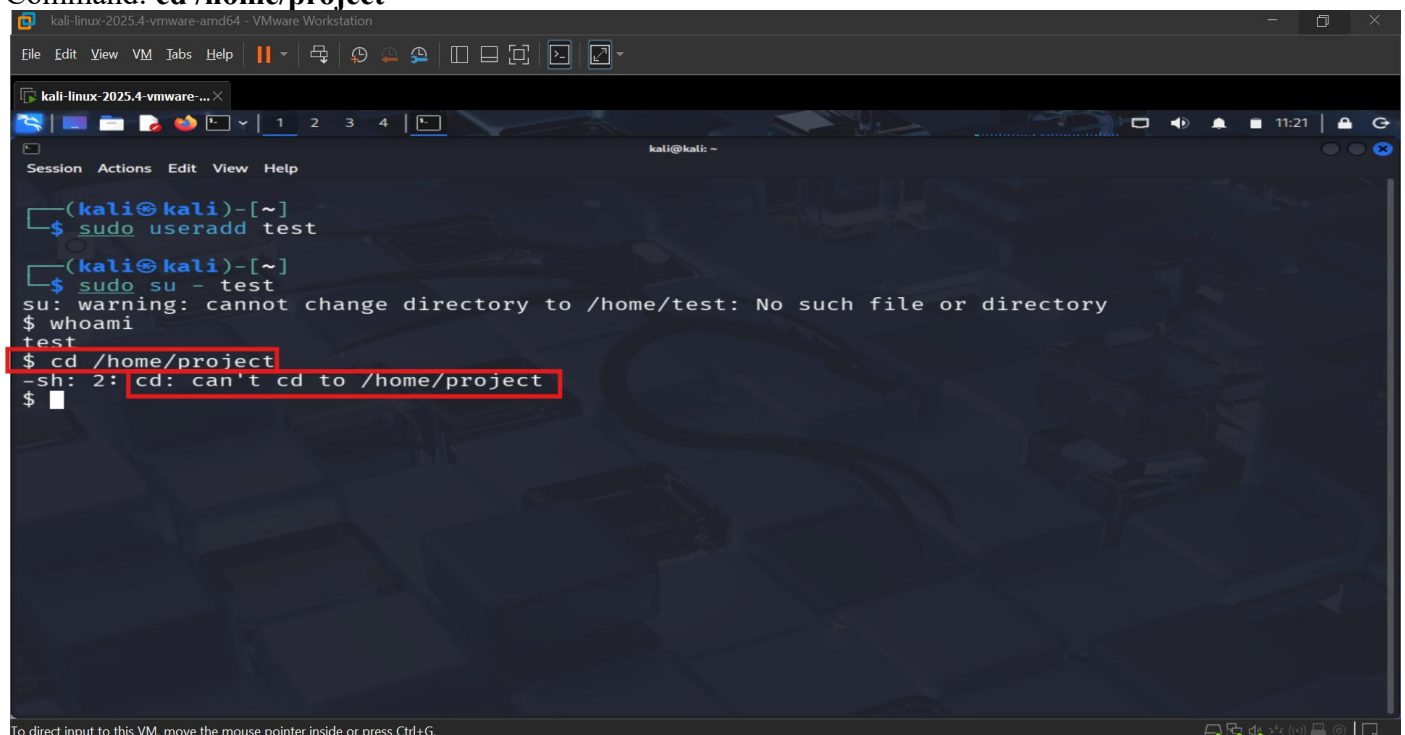
6.2 Unauthorised Access Attempt:

- Create a test user test and try accessing the /home/project directory. We are assuming that the “test” user already exists in the system.
- Command: **sudo useradd test**
sudo su - test



```
kali@kali: ~  
$ sudo useradd test  
$ sudo su - test  
su: warning: cannot change directory to /home/test: No such file or directory  
$
```

Now try to change the directory
Command: **cd /home/project**



```
kali@kali: ~  
$ sudo useradd test  
$ sudo su - test  
su: warning: cannot change directory to /home/test: No such file or directory  
$ whoami  
test  
$ cd /home/project  
-sh: 2: cd: can't cd to /home/project  
$
```

6.3 Testing Command History Configuration

To validate that the command history settings are correctly applied for each user:

- Log in as pa1 or pa5:
- Use command: **history**
- The output should display the last 10 commands.

